

**NOT MEASUREMENT  
SENSITIVE**

**MIL-STD-3005  
20 December 1999**

**DEPARTMENT OF DEFENSE  
TELECOMMUNICATIONS SYSTEMS  
STANDARD**

**ANALOG-TO-DIGITAL CONVERSION  
OF VOICE BY 2,400 BIT/SECOND  
MIXED EXCITATION LINEAR PREDICTION  
(MELP)**



## MIL-STD-3005

### FOREWARD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense (DoD) and is a replacement for FIPSPUB-137, Telecommunications: Analog to Digital Conversion of Voice by 2,400 Bit/Second Linear Predictive Coding.

2. This standard contains design requirements for analog-to-digital (A-D) conversion of voice by 2,400 bit/second Mixed Excitation Linear Prediction (MELP). Adherence to this standard is required to produce interoperable systems at the defined rate and to meet or exceed the minimum performance requirements.

3. Appendix A of this document contains an example of an interoperable MELP algorithm. This information is provided as guidance only.

4. Appendix B contains guidelines for verification of all new implementations of this standard. New implementations must be verified to guarantee that the standard was correctly implemented. This verification process will determine if the standard is interoperable with other MELP implementations and will verify that the performance of the implementation meets or exceeds the performance of the MELP reference coder.

5. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: R224, National Security Agency, 9800 Savage Road STE 6516, Ft. Meade, Maryland 20755-6516 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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## 1. SCOPE

1.1 Scope. This standard establishes interoperability and performance requirements for analog-to-digital (A-D) conversion of voice by 2,400 bit/second Mixed Excitation Linear Prediction (MELP). The requirements presented in this document must be met in order for systems to be interoperable at 2,400 bit/second. Minimum performance requirements are also provided, but may be exceeded. The performance requirements are provided in Appendix B.

## 2. APPLICABLE DOCUMENTS

2.1 General. Documents listed in this section are required in order for the document user to fully understand the guidance being provided by this standard.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

## STANDARDS

## FEDERAL

|              |                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------|
| FED-STD-1016 | Telecommunications: Analog to Digital Conversion of Radio Voice by 4,800 Bit/Second Code Excited Linear Prediction (CELP) |
| FED-STD-1037 | Glossary of Telecommunications Terms                                                                                      |
| FIPSPUB-137  | Telecommunications: Analog to Digital Conversion of Voice by 2,400 Bit/Second Linear Predictive Coding                    |

## MILITARY

|                 |                                                                                        |
|-----------------|----------------------------------------------------------------------------------------|
| MIL-STD-188-113 | Interoperability and Performance Standards for Analog-to-Digital Conversion Techniques |
|-----------------|----------------------------------------------------------------------------------------|

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

(Copies of the Federal Information Processing Standards (FIPS) are available to Department of Defense activities from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Others must request copies of FIPS from the National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161-2171.)

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2.2.2 Other Government documents, drawings, and publication. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

## DoDISS

Department of Defense Index of  
Specifications and Standards

(Copies of the DoDISS are available on a yearly subscription basis either from the Government Printing Office or the DoDSSP Subscription Services, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted should be those listed in the issue of the DoDISS specified in the solicitation. The issues of the documents which have not been adopted should be those in effect on the date of the cited DoDISS.

## NORTH ATLANTIC TREATY ORGANIZATION (NATO)

## STANDARDIZATION AGREEMENT (STANAG's)

## STANAG 4198

Parameters and Coding  
Characteristics That Must be  
Common to Assure Interoperability  
of 2400 BPS Linear Predictive  
Encoded Digital Speech

## STANAG 4209

The NATO Multi-Channel Tactical  
Digital Gateway -- Standards for  
Analogue to Digital Conversion of  
Speech Samples

(Application for copies should be addressed to the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

(Non-Government standards are generally available for reference from libraries. They are also distributed among non-Government standards bodies and using Federal agencies.)

2.4 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard should take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. DEFINITIONS

3.1 Terms. Definitions of terms used in this standard should be as specified in the current edition of FED-STD-1037. In addition, the following definitions are applicable for the purpose of this standard.

3.1.1 Adaptive spectral enhancement. This feature enhances the formant structure of the synthetic speech by use of an adaptive spectral enhancement filter that is applied to the mixed excitation.

3.1.2 Aperiodic pulses. Aperiodic pulses are used in the excitation model of the synthesizer when the aperiodic flag is set to 1. The aperiodic flag is set to one when the jittery voiced state is encountered during the voicing decision process. This feature is used to reduce the buzzy quality of the synthetic speech signal.

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3.1.3 Fourier magnitude modeling. Fourier magnitude modeling involves determining the Fourier magnitudes of the first 10 pitch harmonics of the prediction residual and vector quantizing them with 8 bits for transmission. The use of this technique improves the accuracy of the speech production model at the perceptually important lower frequencies.

3.1.4 Hamming codes. A class of linear codes used for forward error correction. These codes are used only in the unvoiced mode.

3.1.5 Jitter. Random variations introduced into the duration of a signal.

3.1.6 Linear prediction coding. A method for approximating the current speech sample by using a linear combination of past and future speech samples. This method efficiently represents a speech signal and its spectrum characteristics with a very small number of parameters when combined with an appropriate excitation signal.

3.1.7 Mixed excitation. The combination of a periodic function (such as a pulse train) and random noise for use in the excitation model. This combination is applied to sub regions of the frequency domain of the excitation signal.

3.1.8 Prediction coefficients. A set of values that are calculated using a short segment of the input speech signal and provide an estimate of the spectral properties of that signal. These values are determined by performing linear prediction analysis on the input signal. The goal of the analysis is to produce values that minimize the short term mean-squared prediction error over the input segment.

3.1.9 Pulse dispersion. Uses a fixed filter to spread the excitation energy within a pitch period.

3.1.10 Uniform quantizer. A uniform quantizer uses levels and step sizes that are distributed uniformly.

3.1.11 Weighted Euclidean distance. The euclidean distance is a distortion measure between two vectors. In this standard the euclidean distance is determined by summing the squared difference between two vectors for a select number of samples. Normally the euclidean distance is the square root of the measure described in the previous sentence.

3.2 Acronyms used in this standard. The acronyms used in this standard are defined as follows:

A-D - Analog to Digital

DoD - Department of Defense

DoDISS - Department of Defense Index of Specifications and Standards

DoDSSP - Department of Defense Single Stock Point

FEC - Forward Error Correction

LPC - Linear Prediction Coding

LSB - Least Significant Bit

LSF - Line Spectrum Frequency

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MELP - Mixed Excitation Linear Predictions

MSB - Most Significant Bit

MSVQ - Multi-Stage Vector Quantizer

STANAG - Standardization Agreement

#### 4. GENERAL REQUIREMENTS

Not applicable

#### 5. DETAILED REQUIREMENTS

5.1 General. The Mixed Excitation Linear Prediction coder is based on the traditional Linear Prediction Coder (LPC) parametric model, but also includes five additional features. They are mixed excitation, aperiodic pulses, adaptive spectral enhancement, pulse dispersion, and Fourier magnitude modeling. A MELP frame interval is 22.5 ms  $\pm$  0.01 percent in duration and contains 180 voice samples (8,000 samples/second).

5.2 Analog specification. The recommended analog requirements for the MELP coder are for a nominal bandwidth ranging from 100 Hz to 3800 Hz. Although the MELP coder will operate with a more band limited signal, performance degradation will result. To ensure proper operation of the MELP coder, the A-D conversion process should produce peak values of (or near) -32768 and 32767. Additionally, the coder should have unity gain, which means that the output speech level should match that of the input speech.

5.3 Parameter quantization and encoding. The MELP parameters which are quantized and transmitted are the final pitch ( $P_3$ ); the bandpass voicing strengths ( $V_{bp_i}$ ,  $i = 1, 2, \dots, 5$ ); the two gain values ( $G_1$  and  $G_2$ ); the linear prediction coefficients ( $a_i$ ,  $i = 1, 2, \dots, 10$ ); the Fourier magnitudes; and the aperiodic flag. The use of the following quantization procedures is required for interoperability among various implementations.

5.3.1 Pitch and overall voicing. The final pitch ( $P_3$ ), and the low band voicing strength ( $V_{bp_1}$ ), are quantized jointly using 7 bits, as follows. If  $V_{bp_1} \leq 0.6$ , then the frame is unvoiced and the all-zero code is sent. Otherwise, the log of  $P_3$  is quantized with a 99-level uniform scalar quantizer (see 5.3.7) ranging from log20 to log160. The resulting index (range 0 to 98) is then mapped to the transmitted 7-bit codeword using the encode/decode values in table I. All 28 codes with Hamming weight of 1 or 2 are reserved for error protection. This table is also used in decoding the 7-bit pitch code to determine if a frame is voiced, unvoiced, or whether a frame erasure is indicated. A frame is determined unvoiced if the pitch code is all zero or has only one bit set. If two bits are set, then a frame erasure is indicated. Otherwise, the voiced mode is used and the pitch index is determined from the received code according to table I.

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**TABLE I. Encode / decode table for pitch and overall voicing parameter.**

| Code | Index    | Code | Index    | Code | Index    | Code | Index   |
|------|----------|------|----------|------|----------|------|---------|
| 0x0  | UNVOICED | 0x20 | UNVOICED | 0x40 | UNVOICED | 0x60 | ERASURE |
| 0x1  | UNVOICED | 0x21 | ERASURE  | 0x41 | ERASURE  | 0x61 | 68      |
| 0x2  | UNVOICED | 0x22 | ERASURE  | 0x42 | ERASURE  | 0x62 | 69      |
| 0x3  | ERASURE  | 0x23 | 16       | 0x43 | 42       | 0x63 | 70      |
| 0x4  | UNVOICED | 0x24 | ERASURE  | 0x44 | ERASURE  | 0x64 | 71      |
| 0x5  | ERASURE  | 0x25 | 17       | 0x45 | 43       | 0x65 | 72      |
| 0x6  | ERASURE  | 0x26 | 18       | 0x46 | 44       | 0x66 | 73      |
| 0x7  | 0        | 0x27 | 19       | 0x47 | 45       | 0x67 | 74      |
| 0x8  | UNVOICED | 0x28 | ERASURE  | 0x48 | ERASURE  | 0x68 | 75      |
| 0x9  | ERASURE  | 0x29 | 20       | 0x49 | 46       | 0x69 | 76      |
| 0xA  | ERASURE  | 0x2A | 21       | 0x4A | 47       | 0x6A | 77      |
| 0xB  | 1        | 0x2B | 22       | 0x4B | 48       | 0x6B | 78      |
| 0x12 | ERASURE  | 0x32 | 28       | 0x52 | 54       | 0x72 | 85      |
| 0x13 | 5        | 0x33 | 29       | 0x53 | 55       | 0x73 | 86      |
| 0x14 | ERASURE  | 0x34 | 30       | 0x54 | 56       | 0x74 | 87      |
| 0x15 | 6        | 0x35 | 31       | 0x55 | 57       | 0x75 | 88      |
| 0x16 | 7        | 0x36 | 32       | 0x56 | 58       | 0x76 | 89      |
| 0x17 | 8        | 0x37 | 33       | 0x57 | 59       | 0x77 | 90      |
| 0x18 | ERASURE  | 0x38 | 34       | 0x58 | 60       | 0x78 | 91      |
| 0x19 | 9        | 0x39 | 35       | 0x59 | 61       | 0x79 | 92      |
| 0x1A | 10       | 0x3A | 36       | 0x5A | 62       | 0x7A | 93      |
| 0x1B | 11       | 0x3B | 37       | 0x5B | 63       | 0x7B | 94      |
| 0x1C | 12       | 0x3C | 38       | 0x5C | 64       | 0x7C | 95      |
| 0x1D | 13       | 0x3D | 39       | 0x5D | 65       | 0x7D | 96      |
| 0x1E | 14       | 0x3E | 40       | 0x5E | 66       | 0x7E | 97      |
| 0x1F | 15       | 0x3F | 41       | 0x5F | 67       | 0x7F | 98      |

5.3.2 Bandpass voicing. When  $V_{bp1} > 0.6$ , the remaining bandpass voicing strengths are quantized to 1 if their value exceeds 0.6, and quantized to 0 otherwise. There is one exception. If the quantized values of  $V_{bp_i}$ ,  $i = 2, 3, 4, 5$  are 0001, respectively, then  $V_{bp_5}$  is quantized to 0. The quantized values are transmitted using 4 bits. When  $V_{bp1} \leq 0.6$ , the bandpass voicing bits are replaced with FEC parity bits.

5.3.3 Gain. Two gain parameters,  $G_1$  and  $G_2$ , are transmitted each frame.  $G_2$  is quantized to 5 bits using a 32-level uniform quantizer ranging from 10.0 to 77.0 dB. The quantizer index is the transmitted codeword.  $G_1$  is quantized to 3 bits using the following adaptive algorithm. This algorithm determines if the frame is a steady state frame or a transition frame. If  $G_2$ , for the current frame, is within 5 dB of  $G_2$  for the previous frame, and  $G_1$  is within 3 dB of the average of  $G_2$  values for the current and previous frames, then the frame is steady-state and a special code (all zero) is sent to indicate that the decoder should set  $G_1$  to the mean of the  $G_2$  values for the current and previous frames. Otherwise, the frame represents a

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transition and  $G_1$  is quantized with a 7-level uniform quantizer ranging from 6 dB below the minimum of the  $G_2$  values for the current and previous frames to 6 dB above the maximum of those  $G_2$  values. The all-zero codeword is sent for steady state frames and a 7-bit uniform quantizer is used for transition frames. In this case, the quantizer index plus 1 is the transmitted codeword. See 5.3.7 for details on the uniform quantizer.

**5.3.4 Linear prediction coefficients.** The linear prediction coefficients are converted into line spectrum frequencies (LSF) and the resulting LSF vector is checked for monotonicity. If the vector is not monotonic it is adjusted accordingly. The LSF vector is also checked for minimum separation of 50 Hz and adjusted accordingly. The resulting LSF vector is then quantized by a multi-stage vector quantizer (MSVQ). The MSVQ codebook consists of four stages whose indices have 7, 6, 6, and 6 bits, respectively. The quantized LSF vector,  $\hat{f}$ , is the sum of the vectors selected by the search process, with one vector selected from each stage. The MSVQ search finds the codebook vector which minimizes the square of the weighted Euclidean distance,  $d^2$ , between the unquantized and quantized LSF vectors:

$$d^2(f, \hat{f}) = \sum_{i=1}^{10} w_i (f_i - \hat{f}_i)^2, \text{ where } w_i = \begin{cases} P(f_i)^{0.3}, & 1 \leq i \leq 8 \\ 0.64P(f_i)^{0.3}, & i = 9 \\ 0.16P(f_i)^{0.3}, & i = 10 \end{cases}, \text{ EQUATION 1,}$$

$f_i$  is the  $i^{\text{th}}$  component of the unquantized LSF vector, and  $P(f_i)$  is the inverse prediction filter power spectrum evaluated at frequency  $f_i$ . The indices of the four vectors are transmitted. The code vectors and corresponding indices are provided in tables IV-VII.

**5.3.5 Fourier magnitudes.** The ten Fourier magnitudes are coded with an 8-bit vector quantizer. The index of the code vector, which minimizes the weighted Euclidean distance between the input and code vectors, is transmitted. The weights are fixed and are given by:

$$w_i = [117 / (25 + 75(1 + 1.4(f_i/1000)^2)^{0.69})]^2, \quad i = 1, 2, \dots, 10, \text{ EQUATION 2,}$$

where  $f_i = 8000i/60$  is the frequency in Hz corresponding to the  $i^{\text{th}}$  harmonic for a default pitch period of 60 samples. The code vectors and corresponding indices are given in table VIII.

**5.3.6 Aperiodic flag.** The aperiodic flag is a single bit, transmitted as is. The aperiodic flag is set to 1 if  $V_{bp1} < 0.5$  and set to 0 otherwise. When set, this flag tells the decoder that the pulse component of the excitation should be aperiodic, rather than periodic.

**5.3.7 Uniform quantization.** The pitch and gain quantization processes employ uniform quantizers which operate as follows. The stepsize for an  $n$ -level quantizer ranging from  $x_1$  to  $x_2$  is  $s = (x_2 - x_1) / (n - 1)$ . The  $n$  quantizer output values are  $x_1 + i \cdot s$ ,  $i = 0, 1, \dots, n-1$ . The threshold values between levels  $i$  and  $i+1$  are  $x_1 + (0.5 + i)s$ ,  $i = 0, 1, \dots, n-2$ . The quantizer produces  $n$  indices,  $0, 1, \dots, n-1$ , which correspond to an increasing value of the parameter being quantized. For example, let  $x_1 = 1$ ,  $x_2 = 7$ , and  $n = 7$ . This gives  $s = 1$ , levels of  $1, 2, \dots, 7$ , and thresholds of  $1.5, 2.5, \dots, 6.5$ . Index 0 is assigned to input values  $x$ , for which  $x \leq 1.5$ ; index 1 is assigned to input values for which  $1.5 \leq x \leq 2.5$ ; etc.

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5.4 Error protection. Forward Error Correction (FEC) is implemented in the unvoiced mode only, when the Fourier magnitudes, bandpass voicing, and jitter bits need not be transmitted. FEC replaces those 13 bits with the parity bits of three Hamming (7,4) codes and one Hamming (8,4) code. These codes protect the first stage LSF index (7 bits) and both gain indices (8 bits); there is one spare information bit, set to 0.

The protected bits are placed into a column vector,  $u$ , which post-multiplies the parity generator matrix to produce the  $n$ -bit parity vector,  $p = [p_0 \ p_1 \ \dots \ p_{n-1}]^T$ , where  $n$  is 3 or 4. The parity generator matrix for the

Hamming (7,4) code is:  $G_{7,4} = \begin{bmatrix} 1101 \\ 1011 \\ 0111 \end{bmatrix}$ . The parity generator matrix for the Hamming (8,4) code is:

$$G_{8,4} = \begin{bmatrix} 1101 \\ 1011 \\ 0111 \\ 1110 \end{bmatrix}.$$

The 4 most significant bits (MSBs) of the first stage LSF index ( $u = [b_6 \ b_5 \ b_4 \ b_3]^T$ ) are protected by the (8,4) code, with the 4 parity bits written to the LSBs of the bandpass voicing index ( $p_0 \ p_1 \ p_2 \ p_3$ ). The remaining 3 bits of the first stage index and the spare bit ( $u = [b_2 \ b_1 \ b_0 \ 0]^T$ ) are protected with 3 parity bits written to the MSB's of the Fourier magnitude index ( $p_0 \ p_1 \ p_2$ ). The 4 MSBs of the second gain index ( $u = [b_4 \ b_3 \ b_2 \ b_1]^T$ ) are protected with 3 parity bits written to the next 3 bits of the Fourier magnitude index ( $p_0 \ p_1 \ p_2$ ). The LSB of the second gain index and the 3 bit first gain index ( $u = [b_0 \ b_2 \ b_1 \ b_0]^T$ ) are protected with 3 parity bits written to the 2 LSBs of the Fourier magnitude index ( $p_0 \ p_1$ ) and the aperiodic flag ( $p_2$ ). The parenthesized groups of parity bits show their placement in the given index, with the right-most bit having the least significance.

5.5 Transmission format. This section provides information on the transmission rate for the coder, the number of bits allocated for each MELP frame and the transmission order for the bits in each MELP frame.

5.5.1 Transmission rate. The transmission rate should be 2,400 bits/s  $\pm 0.01$  percent. Since all frames contain 54 bits, the frame length is 22.5 ms  $\pm 0.01$  percent.

5.5.2 Bit allocation. Table II shows how the 54 bits in an MELP frame are allocated among the parameters.

| TABLE II. MELP bit allocation. |        |          |
|--------------------------------|--------|----------|
| Parameters                     | Voiced | Unvoiced |
| LSF's                          | 25     | 25       |
| Fourier Magnitudes             | 8      | -        |
| Gain (2 per frame)             | 8      | 8        |
| Pitch, overall voicing         | 7      | 7        |
| Bandpass Voicing               | 4      | -        |
| Aperiodic Flag                 | 1      | -        |
| Error Protection               | -      | 13       |
| Sync Bit                       | 1      | 1        |
| Total Bits / 22.5 ms Frame     | 54     | 54       |



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6.1 Intended use. This standard specifies minimum operability and performance characteristics for analog-to-digital conversion by 2,400 bit/second MELP to be used in the design and installation of new communications subsystems and equipment and in authorized upgrading of existing communications subsystems and equipment. This standard is intended to replace FIPSPUB-137.

6.2 Patent notice. The Government has government purpose license rights under the following listed patents for the benefit of manufacturers of the item for the Government or for use in equipment to be delivered to the Government.

Awarded:

Mixed Excitation Linear Prediction with Fractional Pitch, U.S. Patent Number 5,699,477

Signal Quantizer wherein Average Level Replaces Subframe, U.S. Patent Number 5,794,180

Pending:

Multi-Stage Vector Quantization with Efficient Codebook Search

Adaptive Filter and Filtering Method for Low Bit Rate Coding

6.3 Subject term (key word) listing.

2.4 kbps

2400 bps

Analog-to-digital (A-D) conversion

Encoder/decoder, MELP

Linear prediction coefficients

Low rate

MELP

MELP analyzer

MELP synthesizer

Mixed Excitation Linear Prediction (MELP)

Voice compression

Scalar quantization

Speech coding

Speech compression

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Vector quantization

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| TABLE IV. Codebooks used by the LSF multi-stage quantizer for stage 1 (component values are in Hertz). |            |            |            |             |             |             |             |             |             |             |  |  |  |  |  |  |
|--------------------------------------------------------------------------------------------------------|------------|------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--|--|--|--|--|--|
| Index                                                                                                  | Vector     |            |            |             |             |             |             |             |             |             |  |  |  |  |  |  |
| 0x0                                                                                                    | 355.243052 | 492.660028 | 635.385928 | 980.347948  | 1837.261588 | 2144.996488 | 2413.630272 | 2740.714360 | 3093.750680 | 3368.605604 |  |  |  |  |  |  |
| 0x1                                                                                                    | 484.729576 | 640.392668 | 823.554944 | 1338.889972 | 1880.237972 | 2096.149748 | 2435.878020 | 2765.302236 | 3111.357628 | 3418.139924 |  |  |  |  |  |  |
| 0x2                                                                                                    | 436.939032 | 581.990372 | 733.911476 | 1083.060744 | 1722.101828 | 1952.692520 | 2262.632928 | 2665.653132 | 2988.244500 | 3298.130992 |  |  |  |  |  |  |
| 0x3                                                                                                    | 350.087600 | 694.610176 | 944.914844 | 1378.402864 | 1707.342116 | 2041.440136 | 2412.922496 | 2762.516244 | 3162.275016 | 3440.821584 |  |  |  |  |  |  |
| 0x4                                                                                                    | 269.325932 | 388.049792 | 556.748868 | 785.832892  | 1397.348004 | 1900.273176 | 2320.491456 | 2646.296664 | 3000.054600 | 3327.772772 |  |  |  |  |  |  |
| 0x5                                                                                                    | 419.087356 | 620.798580 | 850.207244 | 1116.317652 | 1781.215188 | 2232.204928 | 2499.593164 | 2780.066296 | 3175.151244 | 3383.999004 |  |  |  |  |  |  |
| 0x6                                                                                                    | 358.942452 | 489.979380 | 640.111288 | 1017.602192 | 1632.583372 | 1860.474172 | 2231.257076 | 2530.919888 | 2914.448960 | 3401.586484 |  |  |  |  |  |  |
| 0x7                                                                                                    | 342.968080 | 472.778640 | 726.418032 | 1259.413024 | 1498.007664 | 1891.490164 | 2466.558164 | 2714.287832 | 3160.189944 | 3419.703608 |  |  |  |  |  |  |
| 0x8                                                                                                    | 335.018920 | 454.914768 | 599.867328 | 961.710264  | 1759.619356 | 2105.336600 | 2379.626816 | 2706.222688 | 3035.591784 | 3319.356816 |  |  |  |  |  |  |
| 0x9                                                                                                    | 442.061924 | 553.856784 | 797.270016 | 1305.514352 | 1865.663680 | 2073.685500 | 2435.956032 | 2730.036752 | 3100.078344 | 3403.177804 |  |  |  |  |  |  |
| 0xa                                                                                                    | 374.716320 | 513.719596 | 672.660540 | 1007.318856 | 1678.355252 | 1896.143836 | 2248.388928 | 2580.640296 | 3008.824168 | 3367.792428 |  |  |  |  |  |  |
| 0xb                                                                                                    | 342.836124 | 602.011044 | 806.724744 | 1354.434868 | 1736.211376 | 1954.730524 | 2467.389528 | 2745.001960 | 3164.511124 | 3452.871652 |  |  |  |  |  |  |
| 0xc                                                                                                    | 257.766340 | 414.160796 | 594.258332 | 830.760316  | 1054.375184 | 1916.648492 | 2340.302436 | 2672.872980 | 3199.556716 | 3428.415672 |  |  |  |  |  |  |
| 0xd                                                                                                    | 292.366648 | 429.559640 | 614.610428 | 820.622344  | 1351.086544 | 2055.154672 | 2396.444508 | 2694.685012 | 3094.502340 | 3359.889892 |  |  |  |  |  |  |
| 0xe                                                                                                    | 298.311292 | 453.999408 | 578.280580 | 905.030628  | 1527.175416 | 1719.933116 | 2049.377252 | 2579.202976 | 2946.545800 | 3302.782036 |  |  |  |  |  |  |
| 0xf                                                                                                    | 336.895904 | 482.206380 | 640.568984 | 999.175084  | 1291.378696 | 1532.487800 | 2241.188304 | 2554.378500 | 2930.959524 | 3361.901736 |  |  |  |  |  |  |
| 0x10                                                                                                   | 258.053976 | 440.262800 | 705.940108 | 988.153356  | 1744.921772 | 2026.777644 | 2372.851292 | 2688.188712 | 3118.794236 | 3391.818028 |  |  |  |  |  |  |
| 0x11                                                                                                   | 471.585160 | 620.421768 | 768.090716 | 1082.870284 | 1751.583152 | 1997.883516 | 2293.602428 | 2743.392004 | 3082.614040 | 3332.107592 |  |  |  |  |  |  |
| 0x12                                                                                                   | 366.246760 | 496.800000 | 672.316592 | 1165.549716 | 1447.016836 | 1687.916328 | 2294.844616 | 2547.156980 | 3219.253312 | 3477.200276 |  |  |  |  |  |  |
| 0x13                                                                                                   | 510.944264 | 676.518572 | 826.792700 | 1134.333084 | 1402.617564 | 1643.895904 | 2356.263864 | 2667.139968 | 3011.790276 | 3369.347468 |  |  |  |  |  |  |
| 0x14                                                                                                   | 255.724280 | 387.165300 | 573.545624 | 785.894372  | 1066.265316 | 1757.711896 | 2263.158644 | 2646.989552 | 3135.291852 | 3456.890708 |  |  |  |  |  |  |
| 0x15                                                                                                   | 378.012868 | 501.412328 | 704.314976 | 917.300392  | 1163.199952 | 2029.238448 | 2375.580324 | 2680.752024 | 3173.105664 | 3386.439868 |  |  |  |  |  |  |
| 0x16                                                                                                   | 363.972960 | 495.552612 | 721.780700 | 993.407816  | 1225.042276 | 1462.770692 | 1739.992988 | 2622.551036 | 3204.830068 | 3432.341068 |  |  |  |  |  |  |
| 0x17                                                                                                   | 464.747332 | 611.589924 | 768.577188 | 1014.013508 | 1220.578772 | 1596.409624 | 2440.187768 | 2705.814872 | 3152.642576 | 3428.078368 |  |  |  |  |  |  |

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| TABLE IV. Codebooks used by the LSF multi-stage quantizer for stage 1 (component values are in Hertz) - Continued. |            |            |             |             |             |             |             |             |             |             |  |  |  |  |  |  |
|--------------------------------------------------------------------------------------------------------------------|------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--|--|--|--|--|--|
| Index                                                                                                              | Vector     |            |             |             |             |             |             |             |             |             |  |  |  |  |  |  |
| 0x18                                                                                                               | 302.335364 | 477.008004 | 604.607176  | 926.489164  | 1569.256396 | 1809.041736 | 2089.583652 | 2607.175324 | 3043.296900 | 3268.448912 |  |  |  |  |  |  |
| 0x19                                                                                                               | 430.245116 | 557.748552 | 772.076800  | 1037.343496 | 1262.090300 | 1915.895200 | 2312.176824 | 2583.310112 | 3099.906888 | 3325.009868 |  |  |  |  |  |  |
| 0x1a                                                                                                               | 349.410284 | 560.269912 | 751.892412  | 993.639536  | 1507.132780 | 1755.265904 | 2086.514692 | 2440.080336 | 2858.434696 | 3309.377008 |  |  |  |  |  |  |
| 0x1b                                                                                                               | 427.173948 | 653.323548 | 827.836272  | 1096.531816 | 1318.221064 | 1677.783892 | 2245.503596 | 2519.509312 | 3076.545064 | 3360.555120 |  |  |  |  |  |  |
| 0x1c                                                                                                               | 299.500472 | 447.090812 | 602.425292  | 837.466012  | 1040.742628 | 1548.003748 | 2279.710052 | 2618.330652 | 3105.809148 | 3444.257700 |  |  |  |  |  |  |
| 0x1d                                                                                                               | 328.461056 | 462.874748 | 624.498484  | 865.029756  | 1068.647964 | 1547.774676 | 2473.663648 | 2745.927484 | 3154.435812 | 3441.285648 |  |  |  |  |  |  |
| 0x1e                                                                                                               | 324.703880 | 468.558984 | 661.255136  | 962.651128  | 1182.703040 | 1462.208432 | 1741.390388 | 2450.460364 | 3083.241456 | 3370.025988 |  |  |  |  |  |  |
| 0x1f                                                                                                               | 348.534068 | 497.361908 | 665.076580  | 927.130960  | 1152.398736 | 1384.022008 | 2206.452184 | 2798.957392 | 3143.444020 | 3404.614276 |  |  |  |  |  |  |
| 0x20                                                                                                               | 282.540208 | 399.817980 | 600.125840  | 1360.887924 | 2103.622308 | 2286.514276 | 2588.754368 | 2874.624788 | 3225.367160 | 3467.056508 |  |  |  |  |  |  |
| 0x21                                                                                                               | 412.601996 | 656.294856 | 1113.328984 | 1647.403296 | 2055.375060 | 2312.331804 | 2665.064932 | 2918.390864 | 3223.101592 | 3484.878096 |  |  |  |  |  |  |
| 0x22                                                                                                               | 311.996136 | 424.640628 | 677.403332  | 1577.771676 | 1868.681796 | 2082.825792 | 2435.651552 | 2698.612244 | 3214.556028 | 3409.172052 |  |  |  |  |  |  |
| 0x23                                                                                                               | 419.770844 | 677.164136 | 1019.950432 | 1497.651320 | 1755.500792 | 2131.790972 | 2580.584380 | 2904.392236 | 3308.333156 | 3532.643648 |  |  |  |  |  |  |
| 0x24                                                                                                               | 259.509376 | 385.637168 | 554.562836  | 937.909320  | 1995.858280 | 2321.667736 | 2564.317268 | 2836.416264 | 3141.165228 | 3417.669004 |  |  |  |  |  |  |
| 0x25                                                                                                               | 259.767772 | 371.811012 | 613.365424  | 1265.584772 | 2012.314700 | 2213.325596 | 2549.977904 | 2808.083496 | 3187.912768 | 3452.612372 |  |  |  |  |  |  |
| 0x26                                                                                                               | 226.714780 | 349.493580 | 608.060808  | 1292.857408 | 1544.272080 | 1868.756264 | 2388.450908 | 2647.670600 | 3142.029760 | 3437.458604 |  |  |  |  |  |  |
| 0x27                                                                                                               | 369.138240 | 511.207808 | 754.415676  | 1319.551916 | 1697.700984 | 1916.897192 | 2447.294220 | 2730.763144 | 3177.868008 | 3446.004236 |  |  |  |  |  |  |
| 0x28                                                                                                               | 252.435392 | 381.234476 | 584.820988  | 1159.922460 | 1898.554296 | 2115.347160 | 2430.113732 | 2702.711664 | 3115.573444 | 3405.707176 |  |  |  |  |  |  |
| 0x29                                                                                                               | 319.502340 | 436.349812 | 863.773064  | 1682.395072 | 1961.092860 | 2203.463320 | 2567.750612 | 2839.694116 | 3219.440192 | 3422.168484 |  |  |  |  |  |  |
| 0x2a                                                                                                               | 331.862240 | 446.244828 | 660.918160  | 1327.774516 | 1694.060384 | 1900.696384 | 2279.667660 | 2520.622328 | 3213.492012 | 3446.511648 |  |  |  |  |  |  |
| 0x2b                                                                                                               | 400.106248 | 561.949040 | 934.342308  | 1358.145284 | 1622.604404 | 2013.068024 | 2389.652200 | 2769.070912 | 3130.311280 | 3413.614884 |  |  |  |  |  |  |
| 0x2c                                                                                                               | 242.657020 | 377.248792 | 522.490964  | 846.946680  | 1761.833916 | 2273.529980 | 2545.114128 | 2796.080976 | 3106.364252 | 3351.943212 |  |  |  |  |  |  |
| 0x2d                                                                                                               | 219.575956 | 360.996896 | 548.251128  | 1088.434748 | 1850.457128 | 2070.726688 | 2414.144860 | 2685.428328 | 3128.683452 | 3450.302588 |  |  |  |  |  |  |
| 0x2e                                                                                                               | 267.708968 | 412.078272 | 563.674080  | 1070.274400 | 1466.291016 | 1693.365224 | 2236.434108 | 2558.971412 | 2977.276580 | 3294.566092 |  |  |  |  |  |  |
| 0x2f                                                                                                               | 280.886640 | 416.620632 | 598.407264  | 1135.358508 | 1546.804680 | 1744.766052 | 2447.398760 | 2759.398560 | 3091.181864 | 3426.265432 |  |  |  |  |  |  |

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TABLE IV. Codebooks used by the LSF multi-stage quantizer for stage 1 (component values are in Hertz) - Continued.

| Index | Vector     |            |             |             |             |             |             |             |             |             |      |      |      |      |      |      |
|-------|------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------|------|------|------|------|------|
|       | 0x30       | 0x31       | 0x32        | 0x33        | 0x34        | 0x35        | 0x36        | 0x37        | 0x38        | 0x39        | 0x3a | 0x3b | 0x3c | 0x3d | 0x3e | 0x3f |
| 0x40  | 262.027288 | 382.135568 | 656.530136  | 1474.423620 | 1840.272900 | 2051.403696 | 2404.401200 | 2679.250592 | 3166.215332 | 3389.927400 |      |      |      |      |      |      |
| 0x41  | 261.244876 | 531.776124 | 972.450092  | 1439.603064 | 1863.321332 | 2137.972540 | 2506.207492 | 2793.512032 | 3204.078640 | 3486.612756 |      |      |      |      |      |      |
| 0x42  | 339.674328 | 453.648992 | 751.280204  | 1389.907636 | 1635.730336 | 1865.774876 | 2173.773284 | 2553.385268 | 3065.628208 | 3273.235436 |      |      |      |      |      |      |
| 0x43  | 341.587388 | 661.908516 | 1017.171716 | 1394.366888 | 1663.345632 | 1920.173796 | 2203.118924 | 2548.413908 | 2988.712388 | 3297.386536 |      |      |      |      |      |      |
| 0x44  | 230.146968 | 343.667980 | 551.302016  | 859.858328  | 1580.424940 | 1832.179784 | 2240.921664 | 2617.020472 | 3037.238376 | 3423.404416 |      |      |      |      |      |      |
| 0x45  | 235.039356 | 364.897428 | 689.403960  | 1029.605932 | 1665.887764 | 1948.938724 | 2346.250084 | 2648.456792 | 3069.623524 | 3389.224348 |      |      |      |      |      |      |
| 0x46  | 308.913508 | 449.624692 | 663.772816  | 1145.008812 | 1351.964772 | 1598.142492 | 1852.252004 | 2454.309724 | 2963.855000 | 3252.933440 |      |      |      |      |      |      |
| 0x47  | 257.909564 | 462.387712 | 741.203612  | 1052.100868 | 1358.228760 | 1622.148340 | 2351.586768 | 2675.735168 | 3092.801340 | 3413.616912 |      |      |      |      |      |      |
| 0x48  | 245.750260 | 376.103252 | 575.477096  | 922.024244  | 1670.456784 | 1903.208808 | 2267.835800 | 2608.720112 | 3005.315992 | 3260.555008 |      |      |      |      |      |      |
| 0x49  | 247.920760 | 379.176512 | 646.023984  | 1306.141840 | 1522.275312 | 2009.603136 | 2373.209384 | 2678.976628 | 3272.521600 | 3480.612912 |      |      |      |      |      |      |
| 0x4a  | 252.521964 | 397.542288 | 729.733388  | 1092.573888 | 1541.289920 | 1831.638300 | 2172.991212 | 2432.845540 | 2863.515608 | 3353.231332 |      |      |      |      |      |      |
| 0x4b  | 293.760896 | 453.119144 | 952.543932  | 1252.249068 | 1524.138240 | 1854.489388 | 2240.400708 | 2529.295340 | 2980.645980 | 3332.247948 |      |      |      |      |      |      |
| 0x4c  | 194.884480 | 303.136856 | 524.001072  | 885.984548  | 1473.173076 | 1919.383060 | 2355.834416 | 2617.938748 | 3072.727588 | 3484.182104 |      |      |      |      |      |      |
| 0x4d  | 221.380896 | 339.827248 | 612.828448  | 974.216724  | 1296.283708 | 1802.879000 | 2241.073592 | 2652.630088 | 3095.005464 | 3428.844228 |      |      |      |      |      |      |
| 0x4e  | 346.441260 | 477.961284 | 659.909792  | 1048.696260 | 1321.778992 | 1560.564952 | 1860.835916 | 2137.948484 | 3028.669084 | 3357.727232 |      |      |      |      |      |      |
| 0x4f  | 306.865700 | 452.086676 | 606.287508  | 958.052088  | 1271.102660 | 1499.649856 | 2168.586976 | 2510.451536 | 2942.205140 | 3376.739692 |      |      |      |      |      |      |
| 0x50  | 367.577380 | 588.407212 | 947.277508  | 1401.718148 | 1710.042324 | 2095.772072 | 2530.043540 | 2877.200112 | 3307.829148 | 3539.939676 |      |      |      |      |      |      |
| 0x51  | 528.676400 | 819.913676 | 1280.496720 | 1637.511820 | 2066.765212 | 2453.835716 | 2824.688768 | 3094.627148 | 3371.254420 | 3558.843104 |      |      |      |      |      |      |
| 0x52  | 545.224416 | 669.303732 | 909.509712  | 1447.146740 | 1725.206608 | 1977.595112 | 2404.918460 | 2678.227616 | 3059.974176 | 3374.444120 |      |      |      |      |      |      |
| 0x53  | 502.175120 | 755.642512 | 1138.330556 | 1488.741484 | 1860.131964 | 2268.931212 | 2653.297860 | 2991.633600 | 3339.490956 | 3557.739556 |      |      |      |      |      |      |
| 0x54  | 298.315792 | 427.893476 | 578.377140  | 1139.800860 | 1526.635152 | 1746.979136 | 2250.458972 | 2556.079216 | 2919.105144 | 3394.443520 |      |      |      |      |      |      |
| 0x55  | 265.345712 | 679.203144 | 917.211964  | 1262.558816 | 1777.907428 | 2105.336432 | 2492.856668 | 2845.667064 | 3222.134752 | 3475.157160 |      |      |      |      |      |      |
| 0x56  | 459.013600 | 591.152844 | 773.066500  | 1211.262108 | 1477.162912 | 1731.845324 | 2117.279752 | 2387.897856 | 3042.563168 | 3318.468632 |      |      |      |      |      |      |
| 0x57  | 465.500052 | 607.671140 | 777.227012  | 1332.425448 | 1605.648308 | 1856.328328 | 2352.959464 | 2590.842552 | 3102.572804 | 3381.365096 |      |      |      |      |      |      |
| 0x58  | 362.053116 | 490.137184 | 745.960236  | 1269.985928 | 1492.284812 | 2059.254896 | 2441.930800 | 2750.527340 | 3227.946188 | 3436.200692 |      |      |      |      |      |      |

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TABLE IV. Codebooks used by the LSF multi-stage quantizer for stage 1 (component values are in Hertz) - Continued.

| Index | Vector     |            |             |             |             |             |             |             |             |             |            |            |            |             |             |
|-------|------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------|------------|------------|-------------|-------------|
|       | 421.179164 | 630.471988 | 1028.686908 | 1430.931296 | 1863.460944 | 2272.510540 | 2652.559160 | 2934.399884 | 3308.053440 | 3523.167936 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x4a  | 483.431132 | 665.740556 | 914.168828  | 1187.178920 | 1730.056668 | 2160.592152 | 2538.107676 | 2833.483468 | 3214.963528 | 3447.815624 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x4b  | 251.069756 | 400.132356 | 683.248580  | 1007.969788 | 1282.548200 | 1622.331828 | 2215.831796 | 2558.840008 | 3112.782068 | 3417.389900 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x4c  | 320.443576 | 444.544064 | 670.428836  | 1086.773516 | 1537.545492 | 1759.446916 | 2524.721292 | 2806.092276 | 3163.121480 | 3453.870028 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x4d  | 425.424868 | 566.142816 | 749.368748  | 1166.345352 | 1466.362360 | 1699.247384 | 2074.299292 | 2340.129404 | 2887.294576 | 3185.769968 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x4e  | 322.007664 | 468.376272 | 648.322576  | 1159.563560 | 1396.322072 | 1703.074704 | 2296.020136 | 2519.346720 | 3200.100636 | 3455.035180 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x4f  | 320.644656 | 596.099288 | 959.540408  | 1218.782764 | 1581.902180 | 1962.503884 | 2385.921440 | 2708.652388 | 3210.663204 | 3479.108204 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x50  | 579.432324 | 803.501624 | 1126.062900 | 1427.862412 | 1802.430756 | 2082.734724 | 2515.367492 | 2857.989008 | 3233.182996 | 3504.253028 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x51  | 551.063564 | 710.286864 | 942.023808  | 1248.062556 | 1525.983900 | 1786.259580 | 2161.901320 | 2562.831384 | 3071.864428 | 3363.444284 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x52  | 609.760980 | 780.872944 | 980.693916  | 1270.962256 | 1550.882860 | 1844.290896 | 2341.687584 | 2627.830008 | 3095.457540 | 3424.026632 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x53  | 368.966568 | 542.830728 | 697.958024  | 962.227744  | 1185.370252 | 1618.496044 | 2046.085300 | 2329.287928 | 2995.799936 | 3309.116368 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x54  | 404.026008 | 558.097656 | 811.595992  | 1010.628240 | 1461.065508 | 2015.035896 | 2351.943240 | 2684.544988 | 3120.826572 | 3372.178544 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x55  | 467.432068 | 604.514196 | 811.435040  | 1070.217980 | 1340.694896 | 1626.960300 | 1978.396652 | 2451.284304 | 2871.931440 | 3178.879984 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x56  | 483.238296 | 618.742476 | 778.447980  | 1096.796044 | 1376.987232 | 1594.994712 | 2291.458884 | 2625.138056 | 2974.698368 | 3369.527960 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x57  | 379.418368 | 521.726544 | 730.197388  | 1034.923576 | 1254.589772 | 1880.551824 | 2225.600464 | 2536.139916 | 3027.365712 | 3270.175404 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x58  | 484.397464 | 617.122684 | 839.871000  | 1142.679472 | 1409.236340 | 2002.964060 | 2395.110708 | 2680.156916 | 3127.463592 | 3368.707032 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x59  | 379.080976 | 610.318280 | 777.690424  | 1032.213916 | 1526.638640 | 1755.544788 | 2096.849712 | 2563.884348 | 2924.848488 | 3252.445700 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x5a  | 519.918808 | 686.437444 | 889.952740  | 1122.475300 | 1354.885456 | 1721.585208 | 2279.554208 | 2577.343264 | 3148.668212 | 3417.217104 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x5b  | 327.841264 | 480.902284 | 620.889844  | 893.219008  | 1114.708076 | 1389.494440 | 2100.358500 | 2641.157268 | 3058.513220 | 3378.730548 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x5c  | 355.124124 | 483.658108 | 677.123732  | 879.520776  | 1104.300568 | 1866.184292 | 2279.529380 | 2581.821656 | 3144.676444 | 3369.651800 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x5d  | 394.104452 | 587.768788 | 783.373896  | 1038.643028 | 1318.632312 | 1564.741340 | 2021.770288 | 2385.037484 | 2817.015404 | 3289.634680 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x5e  | 419.169056 | 568.314168 | 734.757796  | 982.644060  | 1183.660644 | 1521.688088 | 2411.250204 | 2684.267700 | 3102.943040 | 3407.177224 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x5f  | 217.051472 | 442.467788 | 1167.844336 | 1599.059960 | 1943.541700 | 2342.287000 | 2754.109460 | 3023.786416 | 3368.479160 | 3560.444724 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |
| 0x60  | 548.527208 | 936.006868 | 1550.171484 | 1861.420968 | 2197.263736 | 2455.181020 | 2730.234412 | 2965.074536 | 3243.046748 | 3471.871480 | 429.689288 | 559.663168 | 757.935192 | 1288.938980 | 1578.246504 |

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TABLE IV. Codebooks used by the LSF multi-stage quantizer for stage 1 (component values are in Hertz) - Continued.

| Index | Vector     |            |             |             |             |             |             |             |             |             |            |            |             |             |             |
|-------|------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------|------------|-------------|-------------|-------------|
|       | 429.353204 | 649.857460 | 1241.313404 | 1625.726460 | 1852.896036 | 2164.172340 | 2490.051440 | 2778.301472 | 3199.920908 | 3454.966864 | 429.353204 | 649.857460 | 1241.313404 | 1625.726460 | 1852.896036 |
| 0x62  | 429.353204 | 649.857460 | 1241.313404 | 1625.726460 | 1852.896036 | 2164.172340 | 2490.051440 | 2778.301472 | 3199.920908 | 3454.966864 | 530.200920 | 767.252124 | 1229.137072 | 1718.003588 | 2078.513660 |
| 0x63  | 530.200920 | 767.252124 | 1229.137072 | 1718.003588 | 2078.513660 | 2358.501028 | 2693.879268 | 2963.116860 | 3269.410220 | 3513.980576 | 180.250056 | 286.314476 | 782.009868  | 1256.403520 | 1607.783616 |
| 0x64  | 180.250056 | 286.314476 | 782.009868  | 1256.403520 | 1607.783616 | 1964.381668 | 2428.505580 | 2763.402500 | 3181.404500 | 3497.090480 | 363.046928 | 495.176544 | 932.244352  | 1723.729152 | 2069.873516 |
| 0x65  | 363.046928 | 495.176544 | 932.244352  | 1723.729152 | 2069.873516 | 2266.036024 | 2598.054636 | 2863.095040 | 3218.926356 | 3530.567056 | 275.456260 | 401.919616 | 743.453280  | 1360.345704 | 1599.375664 |
| 0x66  | 275.456260 | 401.919616 | 743.453280  | 1360.345704 | 1599.375664 | 1867.905052 | 2173.127540 | 2584.451656 | 3032.595160 | 3309.266888 | 497.729776 | 703.060684 | 1036.341080 | 1397.418484 | 1838.024488 |
| 0x67  | 497.729776 | 703.060684 | 1036.341080 | 1397.418484 | 1838.024488 | 2064.417060 | 2475.461776 | 2841.367628 | 3176.211216 | 3468.295196 | 218.701004 | 364.654668 | 847.953432  | 1455.778028 | 1822.566468 |
| 0x68  | 218.701004 | 364.654668 | 847.953432  | 1455.778028 | 1822.566468 | 2271.184816 | 2688.881072 | 3009.378188 | 3344.588860 | 3544.795392 | 409.265696 | 722.538580 | 1459.908136 | 1792.828940 | 2125.621812 |
| 0x69  | 409.265696 | 722.538580 | 1459.908136 | 1792.828940 | 2125.621812 | 2424.411352 | 2670.935304 | 2920.454980 | 3237.822100 | 3467.081936 | 458.000836 | 574.728436 | 974.289496  | 1430.359792 | 1689.177736 |
| 0x6a  | 458.000836 | 574.728436 | 974.289496  | 1430.359792 | 1689.177736 | 1976.396512 | 2406.885924 | 2657.533792 | 3115.931004 | 3439.716700 | 522.480632 | 818.682484 | 1332.380228 | 1637.908688 | 1904.114776 |
| 0x6b  | 522.480632 | 818.682484 | 1332.380228 | 1637.908688 | 1904.114776 | 2177.842412 | 2487.380184 | 2815.092956 | 3239.457384 | 3495.550436 | 215.477808 | 328.983084 | 581.950052  | 997.003380  | 1496.216816 |
| 0x6c  | 215.477808 | 328.983084 | 581.950052  | 997.003380  | 1496.216816 | 1965.586408 | 2358.623932 | 2637.005088 | 3023.330960 | 3434.554392 | 211.424108 | 336.110144 | 846.609096  | 1295.560356 | 1629.676036 |
| 0x6d  | 211.424108 | 336.110144 | 846.609096  | 1295.560356 | 1629.676036 | 2078.982732 | 2435.617148 | 2785.006628 | 3211.635028 | 3492.684188 | 301.433204 | 429.696220 | 623.869464  | 1219.478592 | 1673.674336 |
| 0x6e  | 301.433204 | 429.696220 | 623.869464  | 1219.478592 | 1673.674336 | 1872.317560 | 2228.603416 | 2498.196592 | 3095.581436 | 3326.314400 | 227.796312 | 619.235776 | 903.737200  | 1186.101972 | 1486.952480 |
| 0x6f  | 227.796312 | 619.235776 | 903.737200  | 1186.101972 | 1486.952480 | 1881.668384 | 2408.921256 | 2705.309088 | 3149.776640 | 3445.450360 | 246.964648 | 410.663624 | 963.556232  | 1336.930556 | 1805.419368 |
| 0x70  | 246.964648 | 410.663624 | 963.556232  | 1336.930556 | 1805.419368 | 2107.567904 | 2494.256680 | 2767.940768 | 3170.005316 | 3459.535500 | 311.054428 | 696.503640 | 1198.813572 | 1667.222600 | 2009.052656 |
| 0x71  | 311.054428 | 696.503640 | 1198.813572 | 1667.222600 | 2009.052656 | 2402.897624 | 2794.768432 | 3103.142944 | 3356.889572 | 3553.501980 | 376.233832 | 487.344044 | 957.121240  | 1293.941648 | 1542.461412 |
| 0x72  | 376.233832 | 487.344044 | 957.121240  | 1293.941648 | 1542.461412 | 1934.271352 | 2279.849188 | 2603.799128 | 3020.524136 | 3276.772280 | 420.577340 | 782.720780 | 1157.816496 | 1409.439164 | 1730.193544 |
| 0x73  | 420.577340 | 782.720780 | 1157.816496 | 1409.439164 | 1730.193544 | 2036.551884 | 2542.586672 | 2940.837648 | 3295.060216 | 3536.318668 | 227.450880 | 354.217032 | 645.854024  | 1103.949568 | 1457.017632 |
| 0x74  | 227.450880 | 354.217032 | 645.854024  | 1103.949568 | 1457.017632 | 1735.123748 | 2145.927528 | 2434.094168 | 2993.342528 | 3398.840732 | 298.009992 | 505.689304 | 804.054936  | 1230.271608 | 1782.344852 |
| 0x75  | 298.009992 | 505.689304 | 804.054936  | 1230.271608 | 1782.344852 | 2092.537116 | 2531.424824 | 2811.813824 | 3208.324332 | 3479.810704 | 361.837248 | 481.881860 | 754.778976  | 1157.999988 | 1373.736592 |
| 0x76  | 361.837248 | 481.881860 | 754.778976  | 1157.999988 | 1373.736592 | 1603.401296 | 1872.798780 | 2597.556168 | 3080.013080 | 3322.141852 | 363.145448 | 654.603360 | 851.016964  | 1201.750484 | 1524.731340 |
| 0x77  | 363.145448 | 654.603360 | 851.016964  | 1201.750484 | 1524.731340 | 1765.555460 | 2194.217812 | 2663.022320 | 3094.972416 | 3372.444548 | 244.837812 | 378.566316 | 819.184860  | 1063.961972 | 1389.825920 |
| 0x78  | 244.837812 | 378.566316 | 819.184860  | 1063.961972 | 1389.825920 | 1926.440476 | 2268.995564 | 2555.088408 | 3159.662448 | 3442.437796 | 289.878276 | 678.873488 | 1094.283708 | 1378.083076 | 1669.550396 |
| 0x79  | 289.878276 | 678.873488 | 1094.283708 | 1378.083076 | 1669.550396 | 2126.537152 | 2546.927840 | 2938.947780 | 3349.758448 | 3564.281964 | 291.135372 | 603.970104 | 806.647404  | 1146.202408 | 1492.922632 |
| 0x7a  | 291.135372 | 603.970104 | 806.647404  | 1146.202408 | 1492.922632 | 1738.308380 | 2185.345584 | 2532.566400 | 3038.664848 | 3362.383560 |            |            |             |             |             |

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| TABLE IV. Codebooks used by the LSF multi-stage quantizer for stage 1 (component values are in Hertz) - Continued. |            |            |             |             |             |             |             |             |             |             |  |  |  |  |
|--------------------------------------------------------------------------------------------------------------------|------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--|--|--|--|
| Index                                                                                                              | Vector     |            |             |             |             |             |             |             |             |             |  |  |  |  |
| 0x7b                                                                                                               | 477.451948 | 711.119268 | 1120.567768 | 1400.126952 | 1652.650504 | 1928.385468 | 2281.368392 | 2627.063532 | 3117.426876 | 3435.299932 |  |  |  |  |
| 0x7c                                                                                                               | 197.859932 | 317.615912 | 561.566488  | 1004.156772 | 1272.079512 | 1786.734928 | 2269.843440 | 2576.009008 | 3054.133780 | 3435.857468 |  |  |  |  |
| 0x7d                                                                                                               | 209.857268 | 338.088960 | 742.900840  | 1010.365752 | 1329.130948 | 1933.476212 | 2274.220164 | 2550.770696 | 2992.557772 | 3438.759304 |  |  |  |  |
| 0x7e                                                                                                               | 371.540116 | 517.541568 | 699.734220  | 1056.769240 | 1360.004676 | 1581.063852 | 1896.072308 | 2138.187064 | 2982.447132 | 3386.475496 |  |  |  |  |
| 0x7f                                                                                                               | 421.325684 | 565.415952 | 733.196828  | 987.054432  | 1189.015404 | 1594.283500 | 2041.152924 | 2300.535092 | 3176.614820 | 3416.630796 |  |  |  |  |

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| TABLE V. Codebooks used by the LSF multi-stage quantizer for stage 2 (component values are in Hertz). |            |             |             |            |            |             |             |             |             |             |  |  |
|-------------------------------------------------------------------------------------------------------|------------|-------------|-------------|------------|------------|-------------|-------------|-------------|-------------|-------------|--|--|
| Index                                                                                                 | Vector     |             |             |            |            |             |             |             |             |             |  |  |
| 0x0                                                                                                   | -1.083480  | -11.578352  | 2.306316    | -13.996316 | -69.456300 | 37.766424   | -72.555080  | -159.215592 | -339.818812 | -152.118616 |  |  |
| 0x1                                                                                                   | -49.649148 | -46.254812  | -78.096196  | -42.879304 | 87.142444  | 6.807124    | -84.402392  | -169.554840 | 114.687420  | 84.983800   |  |  |
| 0x2                                                                                                   | -64.871376 | -74.238856  | -98.423948  | 73.693940  | 8.272440   | -0.445588   | -69.719076  | -97.613228  | -113.804960 | -72.737376  |  |  |
| 0x3                                                                                                   | -76.670260 | -104.000124 | 162.718804  | 146.267556 | 35.949460  | -54.029504  | -165.439464 | -56.673624  | -30.629936  | -20.631364  |  |  |
| 0x4                                                                                                   | 79.209324  | 69.866840   | 42.538208   | -8.700224  | -41.525928 | -87.881980  | -172.058432 | -257.161664 | -205.789544 | -58.729048  |  |  |
| 0x5                                                                                                   | 46.523288  | 32.708904   | 16.372320   | 87.959672  | 17.893436  | -31.334540  | -158.771560 | -210.683128 | 50.801572   | -1.070176   |  |  |
| 0x6                                                                                                   | -24.884988 | 89.877856   | 60.112892   | -46.721488 | -88.565924 | -123.916356 | -98.926960  | -105.548976 | -49.521476  | -27.839204  |  |  |
| 0x7                                                                                                   | 12.216416  | -10.885964  | 50.445364   | 113.287312 | 44.675968  | 7.914896    | -81.805228  | -145.891432 | -269.429460 | -201.234052 |  |  |
| 0x8                                                                                                   | -18.702440 | -28.966588  | -107.616672 | -54.971216 | 189.132464 | 132.336076  | -7.024808   | -69.746976  | -94.375328  | -6.820516   |  |  |
| 0x9                                                                                                   | -56.894540 | -72.024184  | -52.360740  | 193.625260 | 129.735560 | 62.003412   | -71.477448  | -142.162752 | 64.098872   | 11.162704   |  |  |
| 0xa                                                                                                   | -60.650768 | -78.194956  | -63.935496  | 64.655352  | -25.341792 | 84.625644   | 73.669276   | 12.005084   | 149.501796  | 80.509048   |  |  |
| 0xb                                                                                                   | -15.112784 | -31.196064  | 3.849784    | 159.434312 | 93.375988  | 58.684128   | -41.109156  | 87.189492   | 131.801980  | 56.989420   |  |  |
| 0xc                                                                                                   | 19.254800  | 29.158032   | -18.931012  | -97.893236 | 63.340576  | -10.453596  | -104.093976 | -176.150796 | -211.962572 | -17.986996  |  |  |
| 0xd                                                                                                   | 43.597972  | 19.186704   | 57.392544   | 116.543328 | 78.867444  | 26.069228   | -78.842272  | -162.151528 | 142.039780  | 80.101388   |  |  |
| 0xe                                                                                                   | -80.512424 | 6.911116    | 10.447620   | -20.670668 | 122.874212 | 64.969424   | -20.095012  | -41.500620  | 40.870524   | 27.243612   |  |  |
| 0xf                                                                                                   | 15.133536  | -11.533644  | 91.378952   | 186.566144 | 151.176812 | 106.171932  | -10.561384  | -69.816204  | -158.913692 | -147.982316 |  |  |
| 0x10                                                                                                  | -22.857064 | -45.429420  | 43.667084   | 21.242924  | -51.172664 | 132.138704  | 25.815592   | 40.259908   | -43.876200  | -80.833196  |  |  |
| 0x11                                                                                                  | -34.458148 | -37.767448  | -50.349484  | -21.896640 | 91.649508  | 17.432224   | 65.073496   | 11.976292   | -123.733292 | -107.504528 |  |  |
| 0x12                                                                                                  | -22.611732 | -44.342752  | 65.095116   | 32.444480  | -57.425188 | -5.691424   | 138.519252  | 62.980136   | 71.781844   | 39.682944   |  |  |
| 0x13                                                                                                  | -54.683340 | -52.559476  | 144.402736  | 36.354708  | -66.256948 | -121.447840 | -47.001024  | 19.677400   | 55.941392   | 46.382952   |  |  |
| 0x14                                                                                                  | 76.995752  | 60.130164   | 57.113272   | 18.707744  | -53.600588 | 62.799420   | -40.517780  | -114.388064 | -26.126324  | -73.326320  |  |  |
| 0x15                                                                                                  | 25.816108  | 31.076704   | 124.868068  | 63.559872  | 51.002984  | -27.654288  | -97.466628  | -21.956264  | -41.809916  | -45.055148  |  |  |
| 0x16                                                                                                  | -45.228580 | 102.897240  | 83.947112   | 74.901616  | 49.974708  | 8.687204    | 48.971348   | 6.900972    | -8.644580   | -8.383932   |  |  |
| 0x17                                                                                                  | 61.045120  | 182.798948  | 159.671368  | 59.777160  | 29.855824  | -31.491100  | -95.978064  | -78.175532  | -39.050008  | -13.743948  |  |  |

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| TABLE V. Codebooks used by the LSF multi-stage quantizer for stage 2 (component values are in Hertz) - Continued. |            |             |             |             |             |             |             |             |             |             |  |  |  |  |  |
|-------------------------------------------------------------------------------------------------------------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--|--|--|--|--|
| Index                                                                                                             | Vector     |             |             |             |             |             |             |             |             |             |  |  |  |  |  |
| 0x18                                                                                                              | 49.646432  | 28.451656   | 67.128192   | 0.886752    | -49.083656  | 188.228948  | 106.267192  | 13.855044   | -160.204028 | -171.642896 |  |  |  |  |  |
| 0x19                                                                                                              | -10.121400 | -10.494132  | -51.702072  | -102.624176 | 212.475096  | 189.695604  | 82.285124   | 56.342688   | -4.878576   | -11.235844  |  |  |  |  |  |
| 0x1a                                                                                                              | 36.358408  | 21.779392   | 42.357136   | 2.492964    | -42.223212  | 230.770844  | 160.695752  | 154.000972  | 136.475152  | 53.130348   |  |  |  |  |  |
| 0x1b                                                                                                              | -29.676572 | -39.223244  | -35.661060  | 94.983248   | 59.318312   | 2.700204    | 207.235004  | 137.851652  | 158.257256  | 108.921592  |  |  |  |  |  |
| 0x1c                                                                                                              | 122.140104 | 120.913744  | 108.201320  | 67.247396   | 59.132704   | 40.026908   | 43.540560   | 8.987920    | -51.137704  | -38.635328  |  |  |  |  |  |
| 0x1d                                                                                                              | 72.657632  | 68.934704   | 16.956396   | 25.957480   | 190.086812  | 142.948388  | 53.217716   | 2.736948    | -59.085276  | -22.405936  |  |  |  |  |  |
| 0x1e                                                                                                              | 25.878136  | 32.075092   | -18.051520  | 129.615436  | 41.005184   | 141.842100  | 80.837536   | -5.850756   | 113.953980  | 43.969864   |  |  |  |  |  |
| 0x1f                                                                                                              | 56.950488  | 37.665072   | 132.514492  | 135.379112  | 75.671416   | 138.633300  | 118.390580  | 101.879640  | 101.928352  | 42.930520   |  |  |  |  |  |
| 0x20                                                                                                              | 22.106900  | 21.059164   | -34.232504  | -93.793160  | -155.416992 | -194.590552 | 20.038956   | -23.649856  | -114.881156 | 11.324568   |  |  |  |  |  |
| 0x21                                                                                                              | -26.033500 | -35.961000  | -61.732092  | -162.427040 | -26.287444  | -98.269260  | -141.894620 | 95.126388   | -1.505500   | -7.727060   |  |  |  |  |  |
| 0x22                                                                                                              | -31.289284 | -49.998772  | -67.781776  | -161.027220 | -175.279260 | -225.390404 | -61.840204  | 0.995364    | 16.801380   | 48.424344   |  |  |  |  |  |
| 0x23                                                                                                              | -86.553744 | -108.489556 | -17.654176  | -72.132484  | -7.564812   | 8.047932    | -86.936256  | 27.160092   | 12.540832   | 22.373696   |  |  |  |  |  |
| 0x24                                                                                                              | 17.779300  | 15.339312   | -36.725424  | -8.881500   | -102.852904 | -144.867340 | -94.816980  | -196.287232 | 76.219908   | 45.250160   |  |  |  |  |  |
| 0x25                                                                                                              | -19.692492 | -30.220916  | -29.782548  | 17.903896   | -49.210684  | -116.668528 | -256.575612 | -267.358620 | 139.673908  | 78.124496   |  |  |  |  |  |
| 0x26                                                                                                              | -19.207108 | 4.375076    | 21.642788   | -48.371256  | -77.052304  | -161.254660 | -158.684868 | 250.771480  | 151.906788  | 53.175120   |  |  |  |  |  |
| 0x27                                                                                                              | -16.784044 | -37.622168  | 5.782012    | -19.570792  | -88.589344  | -117.076008 | -236.655112 | -78.102088  | -107.415260 | -79.445552  |  |  |  |  |  |
| 0x28                                                                                                              | -33.759476 | -44.175904  | -74.702140  | -135.044380 | -188.927648 | 7.016552    | 82.302568   | 25.474344   | 76.734852   | 27.800992   |  |  |  |  |  |
| 0x29                                                                                                              | -86.590864 | -120.632148 | -148.796704 | -150.115860 | -18.258660  | -84.054596  | 2.635176    | 37.246548   | 34.580064   | 56.689732   |  |  |  |  |  |
| 0x2a                                                                                                              | -70.539400 | -83.361280  | -55.232632  | -27.094692  | -100.538976 | -172.089768 | 53.281532   | -13.033652  | 9.663312    | 46.571280   |  |  |  |  |  |
| 0x2b                                                                                                              | -51.370904 | -79.509096  | -141.551088 | 1.691600    | 57.508000   | -53.104592  | 20.961464   | 69.160500   | 48.868372   | 37.564824   |  |  |  |  |  |
| 0x2c                                                                                                              | -26.397896 | -22.278824  | -58.177544  | -99.990728  | -151.064636 | -22.789692  | -120.173196 | -151.092736 | -107.237000 | -84.906792  |  |  |  |  |  |
| 0x2d                                                                                                              | 20.239624  | 4.006208    | -28.664352  | -11.823076  | -26.946228  | -66.325064  | -59.575592  | -61.081748  | -37.028000  | -19.087332  |  |  |  |  |  |
| 0x2e                                                                                                              | -32.526320 | 6.534672    | -84.214076  | -41.573904  | -42.894712  | -81.510608  | 52.840208   | -43.398352  | 72.024408   | 68.427864   |  |  |  |  |  |
| 0x2f                                                                                                              | 5.449508   | -14.963136  | -41.741028  | 103.234668  | 14.390108   | -64.423760  | 100.949988  | 29.811640   | -101.778792 | -32.993828  |  |  |  |  |  |

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TABLE V. Codebooks used by the LSF multi-stage quantizer for stage 2 (component values are in Hertz) - Continued.

| Index | Vector     |            |            |             |             |             |             |            |            |            |  |  |  |  |  |
|-------|------------|------------|------------|-------------|-------------|-------------|-------------|------------|------------|------------|--|--|--|--|--|
|       |            |            |            |             |             |             |             |            |            |            |  |  |  |  |  |
| 0x30  | 32.295632  | 18.407260  | -5.894076  | -74.483888  | -152.963476 | -204.044336 | 205.952264  | 219.522920 | 81.562784  | 61.202332  |  |  |  |  |  |
| 0x31  | 18.857140  | 38.485396  | -0.989396  | -55.047384  | -48.776448  | -130.013268 | 107.102480  | 146.292968 | -43.136920 | 12.888356  |  |  |  |  |  |
| 0x32  | -0.859316  | -10.029480 | -5.970772  | -37.158360  | -112.881804 | 25.395820   | 308.705512  | 231.753228 | 148.199332 | 77.118376  |  |  |  |  |  |
| 0x33  | -74.718268 | -7.140804  | 45.716792  | 5.980372    | 66.215188   | 4.203068    | 147.785360  | 193.993040 | 80.089172  | 48.570476  |  |  |  |  |  |
| 0x34  | 66.596548  | 54.383604  | 27.843772  | 2.162152    | -78.289776  | -136.087776 | 83.808688   | -2.700500  | -13.633996 | 6.292336   |  |  |  |  |  |
| 0x35  | 44.048796  | 18.914496  | 31.356536  | 6.002560    | -53.040524  | -33.569576  | -141.531972 | 117.338852 | 58.633508  | 3.317584   |  |  |  |  |  |
| 0x36  | 76.666832  | 98.370332  | 72.228980  | -23.754456  | -51.177288  | -139.149128 | -102.191076 | 116.960600 | -15.327760 | -19.772720 |  |  |  |  |  |
| 0x37  | -24.797384 | 74.266648  | 59.426828  | -45.806184  | 45.466244   | -18.249196  | -27.625136  | 91.983212  | -22.390700 | -14.842720 |  |  |  |  |  |
| 0x38  | 3.038084   | -17.986644 | -39.902620 | -148.611816 | -165.180916 | 177.990356  | 106.139356  | 34.196816  | -54.350296 | -73.378024 |  |  |  |  |  |
| 0x39  | -44.622868 | -69.180132 | -94.306064 | -155.434984 | 79.510788   | 55.263412   | 99.390112   | 160.956712 | 93.086912  | 48.449336  |  |  |  |  |  |
| 0x3a  | 8.528100   | -16.482720 | -20.536552 | -119.745020 | -107.365876 | 343.542148  | 310.162272  | 223.345188 | 122.204876 | 31.527824  |  |  |  |  |  |
| 0x3b  | -74.730600 | -98.813932 | -83.739676 | 156.816160  | 172.775428  | 133.228172  | 117.447612  | 109.584436 | 74.203304  | 39.213180  |  |  |  |  |  |
| 0x3c  | 101.110828 | 100.613404 | 64.945400  | 25.583900   | -51.148460  | 26.378476   | -3.647860   | -4.167064  | 141.475064 | 72.420388  |  |  |  |  |  |
| 0x3d  | 27.756364  | 14.300488  | -39.955176 | -107.269680 | 92.431928   | 47.216472   | -69.001244  | 85.735160  | -57.945236 | -63.784492 |  |  |  |  |  |
| 0x3e  | 71.977736  | 63.828088  | -0.871592  | -13.855768  | 26.229036   | -35.582384  | 133.902408  | 73.158532  | 103.646080 | 78.834900  |  |  |  |  |  |
| 0x3f  | 21.330804  | 14.734456  | 35.994576  | 35.047512   | 175.516392  | 109.080608  | 55.660600   | 48.472556  | 81.337388  | 54.264844  |  |  |  |  |  |

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TABLE VI. Codebooks used by the LSF multi-stage quantizer for stage 3 (component values are in Hertz).

| Index | Vectors     |            |            |             |            |            |             |             |             |             |     |     |     |     |     |     |
|-------|-------------|------------|------------|-------------|------------|------------|-------------|-------------|-------------|-------------|-----|-----|-----|-----|-----|-----|
|       | 0x0         | 0x1        | 0x2        | 0x3         | 0x4        | 0x5        | 0x6         | 0x7         | 0x8         | 0x9         | 0xa | 0xb | 0xc | 0xd | 0xe | 0xf |
|       | 32.980820   | 12.950916  | -77.521608 | 3.457884    | 21.115608  | -18.522344 | 8.594232    | -32.455536  | 79.444108   | 57.096016   |     |     |     |     |     |     |
|       | 20.015652   | 21.871904  | -34.912312 | 79.791640   | 42.794480  | 37.292348  | 53.149448   | -12.294248  | -122.971792 | -61.566184  |     |     |     |     |     |     |
|       | 14.547848   | -5.648276  | 8.130888   | -50.548700  | -72.212136 | 99.232328  | -8.558232   | -91.476160  | 75.606244   | 29.898352   |     |     |     |     |     |     |
|       | 46.408076   | 31.719192  | 13.758460  | 3.196728    | -60.667004 | 96.666596  | 30.653072   | -35.450592  | -54.536580  | -83.284228  |     |     |     |     |     |     |
|       | -6.354040   | -25.223476 | -47.876560 | -60.932144  | 82.686748  | 36.111092  | 64.983628   | 82.850232   | 72.591960   | 48.383168   |     |     |     |     |     |     |
|       | -29.084848  | 25.116264  | -32.724520 | 36.106892   | 14.532280  | -62.664768 | 98.917540   | 40.993596   | 64.825552   | 65.144396   |     |     |     |     |     |     |
|       | -0.834372   | -21.045296 | 5.294624   | -4.912748   | -69.596788 | 125.862880 | 43.944528   | 64.578124   | 72.109888   | 20.414812   |     |     |     |     |     |     |
|       | 25.715576   | 39.154404  | -28.547164 | -23.826392  | -71.987108 | 80.416488  | 61.762952   | 28.096916   | 80.217544   | 25.150576   |     |     |     |     |     |     |
|       | -21.216668  | 0.919060   | -54.469596 | 0.970748    | 47.454988  | 79.039928  | 41.946800   | 23.543096   | -61.798552  | -52.983464  |     |     |     |     |     |     |
|       | -19.268628  | -34.590936 | 16.035816  | 29.048544   | 85.936380  | 66.712128  | 53.866020   | -9.536316   | 46.423792   | 40.015764   |     |     |     |     |     |     |
|       | -14.294308  | -28.466636 | 24.792780  | -69.392596  | 1.677280   | 112.654604 | 35.839992   | -4.488360   | -87.320780  | -64.399692  |     |     |     |     |     |     |
|       | 32.723716   | 17.351404  | 63.382408  | 13.164984   | 7.863884   | 32.270756  | 32.808984   | -40.712860  | -159.488912 | -86.625816  |     |     |     |     |     |     |
|       | 27.919388   | 2.081884   | -40.560004 | -85.742272  | 45.749008  | 26.365512  | -37.336432  | 91.871792   | -15.125316  | -27.295192  |     |     |     |     |     |     |
|       | -103.942848 | 30.868692  | 22.905368  | 3.245652    | -10.864848 | 25.740888  | 32.632356   | 14.937776   | 62.672404   | 36.229100   |     |     |     |     |     |     |
|       | -13.644212  | -33.497264 | 7.802936   | -32.739608  | -68.895376 | -14.978252 | -13.743132  | -67.160192  | -154.790440 | -35.284996  |     |     |     |     |     |     |
|       | -61.220672  | -60.013976 | 103.693816 | 39.152964   | 37.872412  | 55.039632  | 39.596404   | -12.290740  | -19.392204  | -4.557552   |     |     |     |     |     |     |
|       | 32.505808   | 15.583912  | 1.784144   | 53.673772   | 73.051196  | 41.469356  | -64.896556  | -7.048480   | 18.025100   | -10.024028  |     |     |     |     |     |     |
|       | 11.483176   | 5.944724   | 17.110496  | 14.790684   | 6.431540   | -20.200572 | -56.308788  | -134.147400 | -214.746576 | -59.407628  |     |     |     |     |     |     |
|       | 43.566172   | 18.989652  | 55.144036  | -3.929840   | -37.379720 | 61.812216  | -49.526292  | 114.098232  | 69.485776   | 21.475288   |     |     |     |     |     |     |
|       | 37.469540   | 28.346816  | 15.723080  | -28.085444  | -6.623116  | -19.773952 | -66.349236  | -31.002968  | -146.990804 | -120.377684 |     |     |     |     |     |     |
|       | 1.120132    | -4.159908  | -14.437336 | -127.887976 | 77.027304  | 45.315748  | 29.269936   | -15.638068  | 8.620996    | 14.473280   |     |     |     |     |     |     |
|       | 44.768968   | 30.467760  | -6.543172  | -95.704460  | 26.850796  | -15.394560 | -25.588856  | -47.367504  | -2.881176   | 39.420112   |     |     |     |     |     |     |
|       | -10.057328  | 2.399412   | 1.061956   | -65.695228  | 20.687476  | -14.959904 | -117.018512 | 92.386240   | 85.463796   | 24.175728   |     |     |     |     |     |     |
|       | 49.644532   | 65.386288  | 9.684648   | 18.181104   | -14.139504 | -5.888800  | -38.585284  | -27.183620  | 58.865948   | 16.431968   |     |     |     |     |     |     |

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| TABLE VI. Codebooks used by the LSF multi-stage quantizer for stage 3 (component values are in Hertz) - Continued. |            |            |            |            |            |             |            |             |             |             |  |  |  |  |  |
|--------------------------------------------------------------------------------------------------------------------|------------|------------|------------|------------|------------|-------------|------------|-------------|-------------|-------------|--|--|--|--|--|
| Index                                                                                                              | Vectors    |            |            |            |            |             |            |             |             |             |  |  |  |  |  |
| 0x18                                                                                                               | -16.380640 | -34.278140 | 0.022544   | 56.067304  | 46.892096  | 30.245204   | -28.026068 | 93.529428   | -56.789800  | -54.481380  |  |  |  |  |  |
| 0x19                                                                                                               | -4.060036  | -29.094112 | -52.545100 | 79.031108  | 69.252092  | 14.267812   | 9.463936   | -65.250772  | 20.751184   | 37.524948   |  |  |  |  |  |
| 0x1a                                                                                                               | 4.583084   | -24.872664 | 107.840788 | 43.419736  | -1.250280  | -4.026280   | -63.961776 | 21.494468   | -36.229604  | -32.554624  |  |  |  |  |  |
| 0x1b                                                                                                               | 2.165696   | -29.151980 | 54.527780  | 94.168364  | 66.493356  | 36.958832   | 1.594416   | -62.430724  | -87.198932  | -28.883704  |  |  |  |  |  |
| 0x1c                                                                                                               | -11.601520 | -21.077228 | -18.656652 | -63.311928 | 81.936576  | 29.605552   | -50.989424 | -103.844388 | -74.116364  | -25.022808  |  |  |  |  |  |
| 0x1d                                                                                                               | -20.789816 | 23.424964  | 27.131436  | -24.919604 | 70.569748  | 26.853608   | -31.525276 | -87.182472  | 84.545612   | 70.302900   |  |  |  |  |  |
| 0x1e                                                                                                               | -49.936228 | -3.446516  | 40.200364  | -18.647372 | 21.569628  | -58.944872  | -17.371572 | -22.311184  | -101.189676 | -49.805164  |  |  |  |  |  |
| 0x1f                                                                                                               | -57.177432 | 62.058144  | 36.515704  | 55.561188  | 6.520588   | 6.297884    | -13.632288 | -52.826640  | -55.443100  | -54.692008  |  |  |  |  |  |
| 0x20                                                                                                               | 6.249736   | 5.520744   | -30.373784 | -42.147600 | 20.147836  | -39.570052  | 54.020328  | -16.561300  | -203.250932 | 20.377448   |  |  |  |  |  |
| 0x21                                                                                                               | -35.205168 | 5.085324   | -51.336656 | 22.956972  | -28.242884 | -23.552672  | 32.053652  | -17.893740  | -122.073052 | -160.749456 |  |  |  |  |  |
| 0x22                                                                                                               | 24.945192  | 13.164164  | 11.748732  | -76.478132 | -4.066148  | -22.276580  | 79.811928  | 50.662004   | -81.866692  | -54.017108  |  |  |  |  |  |
| 0x23                                                                                                               | 40.838864  | 33.272060  | 3.666024   | 12.769580  | -45.868448 | -68.480588  | 92.137976  | 47.209512   | -35.710004  | -18.718076  |  |  |  |  |  |
| 0x24                                                                                                               | -13.862480 | -29.688380 | -16.062832 | -30.689744 | -21.411228 | -15.875376  | 131.557968 | 101.900708  | 23.649376   | 27.497916   |  |  |  |  |  |
| 0x25                                                                                                               | -20.078528 | -17.911912 | -24.694752 | -20.037984 | -59.345372 | -122.612400 | 71.623648  | 21.364328   | 43.982568   | 41.915580   |  |  |  |  |  |
| 0x26                                                                                                               | 13.955236  | -13.670672 | -0.176880  | -47.697124 | -55.913972 | -56.124192  | 62.862572  | -2.427248   | 142.154476  | 104.877084  |  |  |  |  |  |
| 0x27                                                                                                               | 40.271076  | 23.012344  | 27.510152  | -41.497056 | -55.530680 | -58.212060  | 8.071216   | -61.794624  | 51.734348   | 33.354252   |  |  |  |  |  |
| 0x28                                                                                                               | 8.497868   | -2.606700  | -30.219332 | 78.619552  | 30.672148  | 61.511100   | 30.235552  | 110.460236  | 100.718520  | 53.177352   |  |  |  |  |  |
| 0x29                                                                                                               | -20.167520 | -14.445504 | -9.336176  | 127.582768 | -4.294328  | -27.401844  | 61.786412  | 21.701036   | 22.223068   | 15.230212   |  |  |  |  |  |
| 0x2a                                                                                                               | -1.647608  | 54.368956  | 37.237244  | -17.980500 | 42.437252  | 27.778416   | 60.653436  | 59.946944   | 14.306756   | -5.660356   |  |  |  |  |  |
| 0x2b                                                                                                               | 13.352932  | -14.668488 | 56.844876  | 39.456184  | 25.357476  | 11.232768   | 96.450116  | 63.098048   | -28.823600  | -11.658464  |  |  |  |  |  |
| 0x2c                                                                                                               | 16.214176  | 14.498636  | -9.273052  | -9.045960  | -27.283232 | -48.963732  | 39.335716  | 193.888504  | 87.750712   | 40.073856   |  |  |  |  |  |
| 0x2d                                                                                                               | -47.007404 | -53.423716 | 9.553404   | 0.589152   | -50.564088 | -1.666356   | -7.432612  | 56.997036   | 53.725740   | 20.053884   |  |  |  |  |  |
| 0x2e                                                                                                               | -36.631948 | 37.397136  | 27.393352  | -30.540160 | -93.368500 | -20.053984  | -17.314104 | -0.911252   | 14.602000   | 0.825528    |  |  |  |  |  |
| 0x2f                                                                                                               | -1.395788  | -13.464588 | 93.489772  | -1.615688  | -30.138632 | -48.064896  | 15.280056  | 66.923052   | 69.265380   | 43.586932   |  |  |  |  |  |

TABLE VI. Codebooks used by the LSF multi-stage quantizer for stage 3 (component values are in Hertz) - Continued.

| Index | Vectors    |            |            |            |            |             |             |             |            |             |  |  |  |  |  |
|-------|------------|------------|------------|------------|------------|-------------|-------------|-------------|------------|-------------|--|--|--|--|--|
|       |            |            |            |            |            |             |             |             |            |             |  |  |  |  |  |
| 0x30  | 10.569172  | 24.809296  | -60.304192 | -48.704888 | -38.084248 | -30.716660  | -41.121760  | -46.544012  | -33.652504 | -45.618716  |  |  |  |  |  |
| 0x31  | 14.329408  | 2.787968   | 1.889624   | 17.819504  | -36.893736 | -131.985160 | -0.567740   | -55.884160  | -72.492488 | -31.240228  |  |  |  |  |  |
| 0x32  | 27.387340  | 7.644616   | 27.150636  | -5.574300  | -36.398996 | -76.648084  | -129.163304 | 86.750668   | 45.036408  | 6.367308    |  |  |  |  |  |
| 0x33  | 51.258220  | 25.407856  | 15.030632  | 72.681664  | -0.275320  | -33.546500  | -44.555092  | 67.260580   | -40.684228 | -43.996440  |  |  |  |  |  |
| 0x34  | -25.149456 | -55.373592 | -20.508880 | -59.104064 | 39.694332  | -29.855136  | 23.338884   | -29.142064  | -15.073252 | 16.158980   |  |  |  |  |  |
| 0x35  | -24.028792 | -33.787436 | 7.601280   | -28.737668 | -9.413580  | -62.113404  | -56.180560  | -121.529172 | 83.351164  | 60.314356   |  |  |  |  |  |
| 0x36  | 27.417444  | 7.533384   | 0.415240   | -4.640872  | -21.214400 | -41.096728  | -104.321976 | -151.643732 | 171.809252 | 110.383744  |  |  |  |  |  |
| 0x37  | 34.088864  | 14.909740  | 58.394348  | 39.166456  | 2.694692   | -32.094636  | -33.619456  | -101.999952 | 32.469464  | 32.881976   |  |  |  |  |  |
| 0x38  | -13.082648 | -30.073608 | -85.362912 | 6.847348   | -46.908688 | 36.769352   | 43.168012   | 8.622884    | -8.777696  | -8.067804   |  |  |  |  |  |
| 0x39  | -41.948776 | 1.387452   | -52.071212 | 73.038768  | 4.876640   | -17.894024  | -51.727328  | -70.581328  | 17.584816  | -10.379668  |  |  |  |  |  |
| 0x3a  | -12.247768 | -11.237092 | -44.331148 | 30.028040  | -39.230748 | -2.865544   | -119.814076 | 72.574860   | -49.888108 | -64.795236  |  |  |  |  |  |
| 0x3b  | 8.651028   | -16.327560 | 0.858176   | 49.272304  | -33.946904 | 45.591748   | -8.960968   | -67.881588  | -94.941176 | -110.256420 |  |  |  |  |  |
| 0x3c  | -12.528140 | -28.219792 | -66.117904 | 19.523384  | 2.767992   | -68.743624  | -86.900348  | 91.572472   | 64.267052  | 42.045840   |  |  |  |  |  |
| 0x3d  | -18.846356 | -40.824100 | -12.869724 | 6.287764   | -18.501512 | 23.048256   | -60.021784  | -20.368256  | 222.374712 | 136.273740  |  |  |  |  |  |
| 0x3e  | -65.678876 | 21.048360  | 10.027512  | 27.274816  | 12.681548  | -38.586556  | -102.199520 | 98.262288   | 55.721236  | 13.438876   |  |  |  |  |  |
| 0x3f  | 30.881868  | 20.481092  | 5.767264   | 58.983840  | 24.991944  | 11.221364   | 42.022696   | -42.025784  | 160.539288 | 108.165892  |  |  |  |  |  |

**TABLE VII. Codebooks used by the LSF multi-stage quantizer for stage 4 (component values are in Hertz).**

| Index | Vectors    |            |            |            |            |            |            |            |             |            |
|-------|------------|------------|------------|------------|------------|------------|------------|------------|-------------|------------|
| 0x0   | 13.864392  | -19.737336 | 20.320452  | -1.373732  | -24.025380 | 55.403468  | -33.427036 | -7.164116  | -76.739792  | -66.680096 |
| 0x1   | -22.288548 | -34.188684 | 20.704048  | -14.170992 | 28.366312  | -25.985648 | -43.561588 | 15.943264  | 46.563540   | 34.635664  |
| 0x2   | -9.949592  | 14.031280  | 2.022688   | 61.570748  | 11.537028  | -20.475192 | -56.247088 | 12.789900  | -24.688080  | -20.578288 |
| 0x3   | -64.483988 | -2.641932  | 9.354576   | -1.695224  | -11.159228 | 1.977040   | 3.499672   | 18.992872  | -28.880704  | -17.282880 |
| 0x4   | 1.859316   | -18.637868 | 39.190940  | 19.354980  | 44.978516  | 5.672544   | -3.360900  | 30.381176  | -52.519360  | -41.922064 |
| 0x5   | 22.329840  | -8.827276  | 35.918016  | -26.647964 | 26.134024  | -41.064820 | 4.758780   | -49.390996 | 7.984100    | 30.442724  |
| 0x6   | 17.873364  | -3.982856  | 44.497744  | 39.808760  | -3.764812  | -8.364812  | -35.478012 | -11.361596 | 80.956140   | 54.743676  |
| 0x7   | 7.958560   | -32.041208 | 17.138880  | 37.115176  | 17.563628  | -11.659032 | 33.656416  | -18.415836 | 59.808248   | 36.363976  |
| 0x8   | -12.145288 | -24.479824 | 56.629472  | -11.682896 | -31.763652 | 3.683420   | -20.414980 | -31.909604 | 9.772540    | -1.279444  |
| 0x9   | -14.054776 | -35.296828 | -19.199396 | 17.575908  | -28.405600 | -4.153392  | 7.303092   | -29.658076 | -44.023168  | -79.546640 |
| 0xa   | 11.007008  | 19.022892  | 28.676476  | 8.685068   | 41.181528  | 9.850048   | -35.909068 | -57.330104 | 7.240120    | 18.847528  |
| 0xb   | -11.754804 | -0.200860  | -0.397848  | 16.824616  | 41.821960  | 15.431596  | -17.873084 | -46.166628 | -113.237564 | 3.930596   |
| 0xc   | 4.417632   | -13.858636 | 25.993272  | 3.621776   | -13.282620 | -54.866568 | -25.740860 | -8.207356  | -102.337520 | -90.469400 |
| 0xd   | 30.472892  | 2.402004   | -8.414600  | -14.939776 | 20.846288  | -42.379776 | 44.159640  | -1.039776  | -86.199308  | -29.886620 |
| 0xe   | 20.803664  | -5.884716  | -10.076072 | 34.678792  | -1.836392  | -5.648408  | -57.621056 | -64.743808 | -35.249396  | -36.447272 |
| 0xf   | 5.240548   | -7.730720  | -42.546312 | 25.064936  | 15.738880  | 6.281712   | 21.895068  | 5.559424   | -104.670464 | -52.286052 |
| 0x10  | 24.895620  | 3.387108   | 14.067512  | -35.964624 | 34.713716  | 56.939140  | 17.696072  | -3.026896  | -64.985692  | -46.711140 |
| 0x11  | -20.307876 | 32.490736  | -8.269384  | -37.426408 | 15.369516  | 5.518868   | -40.295176 | 49.790088  | -30.418412  | -28.439284 |
| 0x12  | 3.301884   | 23.180704  | -24.454116 | -5.660896  | -46.692200 | 72.857908  | 3.808256   | -13.518736 | -27.566972  | -39.098900 |
| 0x13  | -36.148208 | 33.906036  | -10.643360 | 27.813232  | 7.439476   | 23.088892  | 41.102328  | -5.758800  | -57.094056  | -38.619512 |
| 0x14  | 41.485516  | 33.976276  | 16.557244  | 11.715616  | 1.168420   | 11.215132  | 2.419500   | -2.189148  | -20.660864  | -20.782992 |
| 0x15  | -34.868980 | 44.335072  | 18.371036  | -9.576508  | 6.842808   | -28.376484 | -20.193584 | -36.100756 | -56.015728  | -32.668848 |
| 0x16  | 22.181968  | -4.747624  | -6.587840  | 19.194412  | 13.324592  | 50.283068  | 34.604528  | -63.345460 | -24.597824  | 18.117740  |
| 0x17  | -23.391532 | 11.590768  | 37.601468  | 12.769832  | -9.503604  | -11.279292 | 54.145876  | -50.561068 | 11.992376   | 13.849812  |

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| TABLE VII. Codebooks used by the LSF multi-stage quantizer for stage 4 (component values are in Hertz) - Continued. |            |            |            |            |            |            |            |            |            |            |            |            |  |  |  |  |
|---------------------------------------------------------------------------------------------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|--|--|--|--|
| Index                                                                                                               | Vectors    |            |            |            |            |            |            |            |            |            |            |            |  |  |  |  |
| 0x18                                                                                                                | -7.408196  | -30.925108 | 18.743584  | -54.589100 | 40.004624  | 22.577360  | 8.870712   | -11.437640 | 1.800908   | -11.437640 | 1.800908   | 16.894572  |  |  |  |  |
| 0x19                                                                                                                | 2.106444   | -17.324588 | -22.842364 | -28.871344 | -21.276340 | -13.394672 | 23.784676  | 70.579396  | -37.411672 | 70.579396  | -37.411672 | -52.086276 |  |  |  |  |
| 0x1a                                                                                                                | 15.241316  | 17.603468  | -41.962316 | -4.684472  | 53.083624  | 16.845052  | -13.084284 | -12.831660 | -7.038208  | -12.831660 | -7.038208  | -2.189224  |  |  |  |  |
| 0x1b                                                                                                                | -5.009304  | -10.205264 | -23.680948 | -23.020348 | 55.499976  | 26.375932  | 64.954384  | 7.736592   | -5.772604  | 7.736592   | -5.772604  | -18.452616 |  |  |  |  |
| 0x1c                                                                                                                | 17.559028  | 9.416808   | 39.605736  | 23.848632  | 1.871332   | -54.038420 | 27.998276  | 35.575168  | -29.006648 | 35.575168  | -29.006648 | -1.178764  |  |  |  |  |
| 0x1d                                                                                                                | -6.962476  | 4.105532   | 19.244696  | -23.883640 | -39.505700 | 19.590216  | 53.142492  | 8.583808   | -84.255052 | 8.583808   | -84.255052 | -13.993440 |  |  |  |  |
| 0x1e                                                                                                                | 29.861688  | 7.994612   | -19.893828 | 45.102624  | 8.184900   | -39.195660 | 30.159276  | -34.246888 | 17.732108  | -34.246888 | 17.732108  | 23.665304  |  |  |  |  |
| 0x1f                                                                                                                | 14.491724  | -5.943524  | -5.773136  | 28.083748  | -36.821584 | -25.838832 | 80.000780  | 12.158860  | -10.060840 | 12.158860  | -10.060840 | -7.431432  |  |  |  |  |
| 0x20                                                                                                                | -5.468828  | -26.673576 | -13.033748 | 15.823448  | 2.773560   | 29.638816  | -77.755704 | 29.586208  | 3.212960   | 29.586208  | 3.212960   | -27.223712 |  |  |  |  |
| 0x21                                                                                                                | 4.192508   | -0.595220  | -9.597436  | -29.569652 | -8.967564  | -15.488152 | -87.442240 | 33.505968  | 81.622340  | 33.505968  | 81.622340  | 43.421084  |  |  |  |  |
| 0x22                                                                                                                | 6.918280   | 5.475584   | -34.010376 | 55.845532  | 3.372104   | 41.049620  | -16.152440 | 1.566632   | 52.853932  | 1.566632   | 52.853932  | 16.039552  |  |  |  |  |
| 0x23                                                                                                                | -38.947288 | 18.104492  | -36.755712 | -14.831836 | 14.532324  | -0.120888  | 0.709208   | -29.737616 | 63.593660  | -29.737616 | 63.593660  | 43.486380  |  |  |  |  |
| 0x24                                                                                                                | 16.827216  | -8.372392  | -4.047592  | 14.480476  | 48.383496  | 8.909460   | -23.019528 | 74.803304  | 51.072416  | 74.803304  | 51.072416  | 26.536640  |  |  |  |  |
| 0x25                                                                                                                | 16.908200  | 3.113504   | -0.409796  | -15.891340 | 30.269736  | -61.639948 | -30.195632 | 59.169420  | -8.384824  | 59.169420  | -8.384824  | -4.557260  |  |  |  |  |
| 0x26                                                                                                                | -19.531304 | -19.663028 | 10.757152  | 54.459716  | -5.494000  | -20.867892 | 18.984844  | 61.646372  | 26.824196  | 61.646372  | 26.824196  | 19.117792  |  |  |  |  |
| 0x27                                                                                                                | -29.128836 | -3.459292  | -9.332620  | 17.339504  | 25.026592  | -59.119656 | 54.056544  | -12.055064 | -21.268108 | -12.055064 | -21.268108 | 2.680656   |  |  |  |  |
| 0x28                                                                                                                | -15.645544 | -21.784092 | -4.418296  | -12.523804 | -16.660484 | 74.213892  | 34.056316  | -0.726300  | 51.928168  | -0.726300  | 51.928168  | 22.675180  |  |  |  |  |
| 0x29                                                                                                                | 15.689516  | -11.155860 | -21.276536 | -14.047236 | -54.493236 | 13.666468  | -30.506112 | -37.156156 | 43.355576  | -37.156156 | 43.355576  | 7.350436   |  |  |  |  |
| 0x2a                                                                                                                | -29.735320 | -13.753276 | 3.548232   | 31.269112  | 39.668368  | 44.830620  | -6.977028  | -35.663752 | 30.555460  | -35.663752 | 30.555460  | 4.388992   |  |  |  |  |
| 0x2b                                                                                                                | -31.082216 | 11.584868  | 8.487652   | -33.036500 | -13.157316 | 50.502976  | -23.188224 | -51.524568 | 0.382960   | -51.524568 | 0.382960   | -5.737796  |  |  |  |  |
| 0x2c                                                                                                                | 17.331036  | -10.261824 | 31.780508  | -23.296888 | 8.928572   | 52.597664  | -44.915272 | -30.676572 | 65.429492  | -30.676572 | 65.429492  | 36.915764  |  |  |  |  |
| 0x2d                                                                                                                | 7.656704   | -12.769280 | -34.288816 | -46.295780 | 7.367804   | 0.185976   | -26.244420 | -33.013772 | -69.086572 | -33.013772 | -69.086572 | -27.984080 |  |  |  |  |
| 0x2e                                                                                                                | -18.013668 | -10.073172 | 3.763892   | 24.780428  | -21.653296 | 4.704220   | -13.278056 | -57.884760 | 122.655536 | -57.884760 | 122.655536 | 78.080140  |  |  |  |  |
| 0x2f                                                                                                                | -7.139380  | -3.507464  | -16.812212 | -10.187820 | 11.070308  | -45.039224 | -26.121504 | -78.043388 | 22.171320  | -78.043388 | 22.171320  | 17.720240  |  |  |  |  |

TABLE VII. Codebooks used by the LSF multi-stage quantizer for stage 4 (component values are in Hertz) - Continued.

| Index | Vectors    |            |            |            |            |            |            |            |            |            |  |  |  |  |  |  |
|-------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|--|--|--|--|--|--|
| 0x30  | -0.038328  | -12.205152 | 2.765344   | 43.550748  | 4.322620   | 64.041088  | 37.213004  | 30.822664  | -49.288732 | -32.218564 |  |  |  |  |  |  |
| 0x31  | -24.910772 | 6.883424   | -12.832696 | 1.845308   | -52.037336 | 32.769556  | -20.558604 | 73.189036  | 38.056292  | 9.735312   |  |  |  |  |  |  |
| 0x32  | -1.617564  | 23.527380  | -13.455208 | 32.733468  | 38.018240  | 5.814052   | 46.106580  | 49.198760  | 6.412352   | 3.359260   |  |  |  |  |  |  |
| 0x33  | -22.066160 | 28.427852  | 24.169128  | 9.870268   | 22.294164  | -14.262904 | -2.228472  | 56.488580  | 74.734164  | 52.285380  |  |  |  |  |  |  |
| 0x34  | 20.856168  | 11.434760  | -17.501460 | 19.356592  | -23.927048 | -10.080176 | -29.157472 | 67.028020  | -72.829380 | -38.168352 |  |  |  |  |  |  |
| 0x35  | 4.532676   | 6.949228   | 30.519480  | -1.335668  | -46.754808 | -16.408320 | -71.551928 | 39.060732  | -5.316548  | -16.951812 |  |  |  |  |  |  |
| 0x36  | -6.563488  | -4.683368  | -44.646952 | 11.974452  | -5.360264  | -32.144792 | 27.619440  | 35.933544  | 80.803460  | 59.884136  |  |  |  |  |  |  |
| 0x37  | -12.079848 | 28.782912  | -5.610228  | 32.414792  | -38.174764 | -53.727672 | -3.846380  | 6.256724   | 29.202688  | 12.668764  |  |  |  |  |  |  |
| 0x38  | 9.128584   | -15.964872 | 40.529996  | -25.097428 | -9.404020  | 33.634748  | 25.128176  | 67.999700  | 15.799964  | -7.397816  |  |  |  |  |  |  |
| 0x39  | -0.584244  | -19.878520 | 15.734836  | -26.676744 | -33.555660 | -67.913680 | 2.189504   | 28.161576  | 19.481028  | 6.669180   |  |  |  |  |  |  |
| 0x3a  | 27.830096  | 3.212196   | -18.081108 | -53.418624 | 16.337124  | 17.790228  | -0.935908  | 18.558948  | 66.264504  | 36.820628  |  |  |  |  |  |  |
| 0x3b  | -8.894760  | 17.362400  | 13.714624  | -55.179844 | 3.355216   | -25.666356 | 52.092692  | 9.626172   | 37.176688  | 36.454116  |  |  |  |  |  |  |
| 0x3c  | 24.708572  | 2.926604   | 19.614548  | 11.151496  | -50.921244 | 31.058220  | 38.814708  | 9.470088   | 51.206496  | 7.702788   |  |  |  |  |  |  |
| 0x3d  | 26.751716  | 10.221092  | 20.453016  | -38.210332 | -40.726020 | -25.178320 | -9.964692  | -19.559148 | -45.512164 | -28.789868 |  |  |  |  |  |  |
| 0x3e  | 19.678856  | 24.642696  | -2.825972  | -14.446076 | -19.540088 | -8.693844  | 1.168576   | -59.036728 | 102.917656 | 79.647188  |  |  |  |  |  |  |
| 0x3f  | -7.553260  | 16.710720  | -31.982072 | -28.903800 | -40.696264 | -28.379732 | 36.020044  | -21.981404 | -5.020000  | -4.84708   |  |  |  |  |  |  |

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| TABLE VIII. Codebook used by the Fourier magnitude vector quantizer |          |          |          |          |          |          |          |          |          |          |     |     |     |     |     |     |
|---------------------------------------------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----|-----|-----|-----|-----|-----|
| Index                                                               | Vectors  |          |          |          |          |          |          |          |          |          |     |     |     |     |     |     |
|                                                                     | 0x0      | 0x1      | 0x2      | 0x3      | 0x4      | 0x5      | 0x6      | 0x7      | 0x8      | 0x9      | 0xa | 0xb | 0xc | 0xd | 0xe | 0xf |
|                                                                     | 0.822998 | 1.496297 | 0.584847 | 1.313507 | 0.846008 | 0.614041 | 0.615298 | 0.881819 | 1.227331 | 0.902116 |     |     |     |     |     |     |
|                                                                     | 1.248150 | 1.020382 | 0.517184 | 1.489079 | 0.650498 | 0.716904 | 0.650349 | 1.260358 | 0.871024 | 0.907932 |     |     |     |     |     |     |
|                                                                     | 1.167392 | 1.468091 | 0.743322 | 0.712255 | 1.564293 | 0.721041 | 0.528277 | 0.657527 | 0.763228 | 0.846655 |     |     |     |     |     |     |
|                                                                     | 1.043080 | 1.570270 | 0.444422 | 0.933200 | 1.211773 | 0.669870 | 0.932675 | 0.806473 | 0.913787 | 0.868244 |     |     |     |     |     |     |
|                                                                     | 0.759126 | 1.037894 | 1.126967 | 1.133186 | 0.789090 | 0.807399 | 1.054906 | 0.949898 | 1.007863 | 1.119437 |     |     |     |     |     |     |
|                                                                     | 0.977225 | 1.130888 | 1.044601 | 1.432985 | 0.819888 | 0.517787 | 0.932401 | 0.964477 | 0.886867 | 0.895436 |     |     |     |     |     |     |
|                                                                     | 0.666187 | 1.364039 | 1.088940 | 0.902131 | 1.110838 | 0.980791 | 0.926721 | 0.867719 | 0.925053 | 0.913544 |     |     |     |     |     |     |
|                                                                     | 0.983185 | 1.460579 | 0.732273 | 0.923475 | 1.088579 | 0.952291 | 0.789664 | 1.025082 | 0.904670 | 0.854338 |     |     |     |     |     |     |
|                                                                     | 0.697275 | 1.651033 | 0.506420 | 0.809603 | 1.113992 | 1.083678 | 0.629739 | 0.986236 | 0.871250 | 0.997090 |     |     |     |     |     |     |
|                                                                     | 0.749416 | 1.354976 | 0.731452 | 1.281438 | 1.080946 | 1.037235 | 0.748186 | 0.821493 | 0.879839 | 0.965350 |     |     |     |     |     |     |
|                                                                     | 0.976897 | 2.009469 | 0.741592 | 0.606862 | 0.644927 | 0.687413 | 0.760360 | 0.833318 | 0.869851 | 0.810534 |     |     |     |     |     |     |
|                                                                     | 0.576891 | 1.527863 | 0.657703 | 1.006739 | 1.226803 | 0.786733 | 0.991212 | 0.821165 | 1.014984 | 0.878058 |     |     |     |     |     |     |
|                                                                     | 0.517211 | 1.220859 | 0.886556 | 0.980153 | 0.884365 | 1.041315 | 0.949276 | 0.881296 | 1.373464 | 0.890748 |     |     |     |     |     |     |
|                                                                     | 0.651809 | 1.106789 | 0.885180 | 1.329281 | 1.124388 | 0.626778 | 1.261132 | 0.822453 | 0.966925 | 0.806183 |     |     |     |     |     |     |
|                                                                     | 0.614335 | 1.430380 | 0.899696 | 0.847168 | 0.624567 | 1.307056 | 1.360741 | 0.735438 | 0.854307 | 0.674763 |     |     |     |     |     |     |
|                                                                     | 0.830604 | 1.373081 | 0.785967 | 0.735951 | 0.968745 | 1.100706 | 0.927619 | 0.728449 | 1.292031 | 0.884959 |     |     |     |     |     |     |
|                                                                     | 1.084085 | 1.385767 | 0.448480 | 1.297458 | 0.770611 | 0.719263 | 1.080766 | 0.909836 | 0.838299 | 0.909742 |     |     |     |     |     |     |
|                                                                     | 1.244608 | 1.158451 | 0.615757 | 1.169599 | 1.032735 | 0.921809 | 0.916509 | 1.060343 | 0.836990 | 0.767101 |     |     |     |     |     |     |
|                                                                     | 1.313869 | 1.341417 | 0.481710 | 0.685601 | 1.086817 | 1.003461 | 0.762127 | 0.832518 | 1.081618 | 0.917168 |     |     |     |     |     |     |
|                                                                     | 1.427137 | 0.932544 | 0.595881 | 0.652610 | 1.073852 | 0.847480 | 1.264058 | 0.899502 | 0.899264 | 0.958964 |     |     |     |     |     |     |
|                                                                     | 1.027745 | 1.214124 | 0.896291 | 0.972030 | 0.841943 | 0.738307 | 1.048254 | 1.077882 | 0.952564 | 1.056319 |     |     |     |     |     |     |
|                                                                     | 0.952252 | 1.098463 | 0.690316 | 1.175721 | 0.911712 | 0.962678 | 0.859269 | 1.138901 | 1.074558 | 0.924091 |     |     |     |     |     |     |
|                                                                     | 0.902314 | 1.163750 | 1.143587 | 0.814860 | 0.923235 | 0.973077 | 1.098871 | 0.966014 | 0.968856 | 0.893133 |     |     |     |     |     |     |
|                                                                     | 1.133879 | 1.110463 | 0.687589 | 0.793794 | 1.026535 | 1.076900 | 1.185281 | 0.821843 | 0.845464 | 1.079782 |     |     |     |     |     |     |

| Index | Vectors  |          |          |          |          |          |          |          |          |          |  |  |  |  |
|-------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|--|--|--|--|
| 0x18  | 0.623910 | 1.269921 | 0.722072 | 0.990256 | 1.183314 | 1.161476 | 0.677381 | 1.199311 | 0.902081 | 0.883634 |  |  |  |  |
| 0x19  | 0.585861 | 1.189163 | 0.882952 | 1.134136 | 0.900218 | 1.263223 | 0.995922 | 1.011126 | 0.906268 | 0.863339 |  |  |  |  |
| 0x1a  | 0.687616 | 1.562003 | 0.815795 | 0.701934 | 0.789752 | 1.204232 | 0.821560 | 1.168157 | 0.888904 | 0.860778 |  |  |  |  |
| 0x1b  | 0.950686 | 1.329037 | 0.692678 | 0.754646 | 1.095995 | 1.446339 | 0.941535 | 0.710369 | 0.789949 | 0.826271 |  |  |  |  |
| 0x1c  | 0.520026 | 1.293024 | 0.938993 | 0.990412 | 0.897114 | 0.914170 | 1.358696 | 0.952011 | 0.936183 | 0.837973 |  |  |  |  |
| 0x1d  | 0.669622 | 1.291638 | 0.749196 | 1.167486 | 1.042641 | 0.721329 | 1.005261 | 1.252692 | 0.883186 | 0.854416 |  |  |  |  |
| 0x1e  | 0.497926 | 1.222508 | 1.403952 | 0.799058 | 0.798533 | 0.793545 | 1.264461 | 0.958949 | 0.843825 | 0.909886 |  |  |  |  |
| 0x1f  | 0.891705 | 1.342163 | 0.761635 | 0.758182 | 0.987397 | 1.008477 | 1.377249 | 0.845635 | 0.845744 | 0.822265 |  |  |  |  |
| 0x20  | 1.174305 | 0.454482 | 0.432532 | 0.691549 | 1.066271 | 0.890505 | 0.845957 | 1.125649 | 1.338280 | 1.204068 |  |  |  |  |
| 0x21  | 1.607432 | 0.688224 | 0.500617 | 1.288348 | 0.678978 | 0.674969 | 1.047764 | 1.012974 | 0.954688 | 0.803209 |  |  |  |  |
| 0x22  | 1.739179 | 0.919542 | 0.639494 | 0.775935 | 1.021945 | 0.991820 | 0.887633 | 0.876449 | 0.796800 | 0.771028 |  |  |  |  |
| 0x23  | 1.620174 | 1.211703 | 0.610139 | 1.372774 | 1.057336 | 0.618126 | 0.642995 | 0.667444 | 0.666172 | 0.676810 |  |  |  |  |
| 0x24  | 1.219952 | 0.844191 | 1.016601 | 0.867402 | 1.006595 | 0.940309 | 0.893858 | 0.963053 | 0.730414 | 1.286290 |  |  |  |  |
| 0x25  | 1.251753 | 1.004762 | 0.777132 | 1.203828 | 0.656382 | 0.657205 | 1.233246 | 0.874799 | 1.106710 | 0.855250 |  |  |  |  |
| 0x26  | 1.348819 | 0.791939 | 1.030755 | 0.765314 | 1.014939 | 1.044297 | 0.747682 | 1.309667 | 0.853913 | 0.723126 |  |  |  |  |
| 0x27  | 1.262759 | 1.109681 | 0.672002 | 1.109957 | 0.827608 | 1.272683 | 0.670943 | 1.068174 | 0.752511 | 0.878203 |  |  |  |  |
| 0x28  | 1.108504 | 1.250712 | 0.620039 | 0.752463 | 1.439216 | 0.890828 | 1.018300 | 0.736766 | 0.910504 | 0.849118 |  |  |  |  |
| 0x29  | 1.222589 | 1.083869 | 0.627894 | 1.150342 | 1.377353 | 0.785771 | 0.679900 | 0.897937 | 0.854521 | 0.920765 |  |  |  |  |
| 0x2a  | 1.326349 | 1.690290 | 1.188837 | 0.595281 | 0.602145 | 0.628852 | 0.676343 | 0.829212 | 0.785923 | 0.799661 |  |  |  |  |
| 0x2b  | 1.499772 | 1.400400 | 0.655881 | 0.892565 | 1.089036 | 0.702486 | 0.835749 | 0.800092 | 0.731263 | 0.835818 |  |  |  |  |
| 0x2c  | 1.126664 | 1.353291 | 0.898265 | 0.897870 | 0.934461 | 1.003119 | 0.982470 | 0.815886 | 0.952342 | 0.831774 |  |  |  |  |
| 0x2d  | 1.316478 | 1.060041 | 0.667653 | 1.065408 | 0.910377 | 1.002121 | 1.339044 | 0.710796 | 0.720291 | 0.813624 |  |  |  |  |
| 0x2e  | 0.976194 | 1.377380 | 1.071262 | 0.789567 | 1.085928 | 1.253806 | 1.017727 | 0.719049 | 0.561583 | 0.692329 |  |  |  |  |
| 0x2f  | 1.202869 | 1.059008 | 0.962883 | 0.999919 | 0.524897 | 1.074531 | 1.091141 | 0.910749 | 1.044517 | 0.859795 |  |  |  |  |

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TABLE VIII. Codebook used by the Fourier magnitude vector quantizer - Continued.

| Index | Vectors  |          |          |          |          |          |          |          |          |          |
|-------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
|       | 1.413976 | 0.836644 | 0.666408 | 1.039645 | 1.088087 | 0.820217 | 0.816371 | 1.049252 | 1.083270 | 0.881172 |
| 0x30  | 1.413976 | 0.836644 | 0.666408 | 1.039645 | 1.088087 | 0.820217 | 0.816371 | 1.049252 | 1.083270 | 0.881172 |
| 0x31  | 1.388093 | 0.807110 | 0.781561 | 0.964242 | 0.973557 | 1.138037 | 1.026733 | 0.970914 | 0.866461 | 0.841057 |
| 0x32  | 1.456077 | 0.982162 | 0.557993 | 0.738683 | 1.375522 | 0.772422 | 0.644426 | 0.952329 | 0.954631 | 1.004692 |
| 0x33  | 1.606208 | 1.220452 | 0.738208 | 0.744965 | 0.659231 | 0.813191 | 0.932141 | 0.922849 | 0.956161 | 0.892080 |
| 0x34  | 1.126839 | 1.038057 | 0.935449 | 1.012686 | 1.080640 | 0.884338 | 1.126468 | 1.076987 | 0.817224 | 0.736383 |
| 0x35  | 1.215351 | 0.845602 | 0.829299 | 1.204972 | 0.891515 | 0.934431 | 0.887728 | 0.938925 | 0.955741 | 1.114145 |
| 0x36  | 1.206387 | 0.977863 | 1.051199 | 0.847855 | 0.939759 | 1.254109 | 0.970361 | 0.901040 | 0.844070 | 0.819544 |
| 0x37  | 1.207602 | 0.976323 | 0.841307 | 0.870300 | 0.877422 | 0.988163 | 0.931714 | 1.031677 | 1.205409 | 0.903902 |
| 0x38  | 1.095130 | 1.105129 | 0.722843 | 0.943180 | 1.211443 | 1.178861 | 0.634252 | 0.870933 | 1.011727 | 0.936443 |
| 0x39  | 1.032014 | 0.998735 | 0.686262 | 1.100404 | 1.022352 | 1.222401 | 1.146689 | 0.812882 | 0.927129 | 0.801404 |
| 0x3a  | 1.131931 | 1.489739 | 0.677826 | 0.610733 | 0.759219 | 0.808122 | 1.042500 | 1.077132 | 1.035717 | 0.906736 |
| 0x3b  | 1.342504 | 1.076171 | 0.583400 | 0.764890 | 0.818097 | 1.293405 | 1.058200 | 0.852259 | 0.958646 | 0.872888 |
| 0x3c  | 1.308520 | 1.219689 | 0.935572 | 0.714861 | 1.129251 | 0.856944 | 0.920880 | 0.974834 | 0.821443 | 0.847930 |
| 0x3d  | 1.372592 | 0.887155 | 0.710732 | 1.029836 | 1.278897 | 0.650119 | 1.136552 | 0.881858 | 0.850047 | 0.819899 |
| 0x3e  | 1.019956 | 1.381944 | 1.089064 | 0.576825 | 0.594933 | 0.752016 | 1.265619 | 1.129530 | 0.867864 | 0.768122 |
| 0x3f  | 1.154864 | 1.069591 | 0.913967 | 0.796105 | 0.790255 | 0.854206 | 1.523905 | 0.856969 | 0.887946 | 0.756665 |
| 0x40  | 0.874806 | 1.645517 | 0.834275 | 1.578196 | 0.903996 | 0.547038 | 0.565520 | 0.704114 | 0.697290 | 0.713786 |
| 0x41  | 1.153418 | 1.095160 | 0.870564 | 1.655735 | 1.071838 | 0.806210 | 0.607539 | 0.566859 | 0.683404 | 0.826413 |
| 0x42  | 0.830864 | 1.448871 | 1.014508 | 0.939034 | 1.574030 | 0.796490 | 0.654852 | 0.558685 | 0.641596 | 0.797132 |
| 0x43  | 1.125819 | 1.396448 | 0.746324 | 1.295697 | 1.137447 | 1.301677 | 0.574765 | 0.486441 | 0.489204 | 0.608138 |
| 0x44  | 0.832614 | 0.802739 | 1.148272 | 1.316531 | 1.305743 | 0.696667 | 0.940671 | 0.848198 | 0.872696 | 0.851311 |
| 0x45  | 1.260278 | 0.958467 | 0.667246 | 1.406435 | 0.859092 | 0.534003 | 0.485425 | 0.723814 | 1.418320 | 0.944772 |
| 0x46  | 0.887527 | 1.330678 | 1.062704 | 0.552689 | 1.098956 | 0.917567 | 0.874732 | 1.075172 | 0.845710 | 1.019528 |
| 0x47  | 0.906302 | 1.326878 | 0.632127 | 0.781817 | 1.111568 | 0.753606 | 0.896250 | 0.945621 | 0.969063 | 1.304544 |
| 0x48  | 0.163798 | 1.616260 | 0.520967 | 1.244674 | 0.597986 | 1.174228 | 0.802735 | 1.079115 | 0.892800 | 0.877604 |

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TABLE VIII. Codebook used by the Fourier magnitude vector quantizer - Continued.

| Index | Vectors  |          |          |          |          |          |          |          |          |          |      |      |      |      |      |
|-------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|------|------|------|------|------|
|       | 0x49     | 0x4a     | 0x4b     | 0x4c     | 0x4d     | 0x4e     | 0x4f     | 0x50     | 0x51     | 0x52     | 0x53 | 0x54 | 0x55 | 0x56 | 0x57 |
|       | 0.717419 | 1.075445 | 0.997204 | 1.206762 | 1.312107 | 1.316050 | 0.696587 | 0.745413 | 0.739740 | 0.731447 |      |      |      |      |      |
|       | 0.339447 | 1.827655 | 0.684798 | 0.716559 | 0.676084 | 1.081139 | 0.894730 | 0.954454 | 0.948165 | 0.910477 |      |      |      |      |      |
|       | 0.864667 | 1.451421 | 0.725543 | 1.176080 | 1.209637 | 1.083631 | 1.163663 | 0.662386 | 0.501862 | 0.463726 |      |      |      |      |      |
|       | 0.563978 | 1.135516 | 0.835026 | 1.736364 | 1.000137 | 0.805360 | 0.828496 | 0.833563 | 0.776277 | 0.805186 |      |      |      |      |      |
|       | 1.009051 | 1.360317 | 0.668259 | 1.417717 | 1.518623 | 0.638014 | 0.446922 | 0.496800 | 0.693633 | 0.726920 |      |      |      |      |      |
|       | 0.186440 | 1.625134 | 0.835253 | 1.175778 | 0.860628 | 0.885086 | 1.011866 | 0.789303 | 1.010232 | 0.802450 |      |      |      |      |      |
|       | 0.652311 | 1.599127 | 0.599516 | 1.058215 | 0.801753 | 0.992815 | 1.146092 | 0.829615 | 0.939228 | 0.837653 |      |      |      |      |      |
|       | 0.759780 | 1.399480 | 0.987329 | 1.024824 | 0.974458 | 0.551246 | 1.150852 | 0.844956 | 1.016771 | 0.925568 |      |      |      |      |      |
|       | 1.069814 | 1.160678 | 0.803250 | 1.193140 | 1.112463 | 0.744748 | 1.070724 | 0.739996 | 0.924675 | 0.942765 |      |      |      |      |      |
|       | 1.168517 | 1.633441 | 0.774778 | 0.736824 | 1.089258 | 1.057042 | 0.783527 | 0.691403 | 0.779568 | 0.728487 |      |      |      |      |      |
|       | 1.301817 | 1.572909 | 0.526128 | 0.558027 | 0.660410 | 1.078262 | 1.089774 | 0.718543 | 0.753433 | 0.965337 |      |      |      |      |      |
|       | 0.891795 | 1.262297 | 0.911545 | 1.214103 | 0.776010 | 0.944986 | 1.138773 | 0.772798 | 0.957464 | 0.885334 |      |      |      |      |      |
|       | 1.219015 | 1.048683 | 0.943178 | 1.150567 | 0.837824 | 0.566225 | 0.660764 | 1.219023 | 1.019092 | 0.988723 |      |      |      |      |      |
|       | 0.925533 | 1.350162 | 1.164350 | 0.542695 | 1.227640 | 0.600203 | 1.205779 | 0.701896 | 0.932956 | 0.815466 |      |      |      |      |      |
|       | 1.019130 | 1.292473 | 0.796847 | 0.819714 | 1.254208 | 0.522062 | 1.293349 | 0.823616 | 0.832862 | 0.890015 |      |      |      |      |      |
|       | 0.686388 | 1.534513 | 1.023273 | 1.065238 | 0.769952 | 0.914897 | 0.958226 | 0.965983 | 0.847693 | 0.876086 |      |      |      |      |      |
|       | 0.560425 | 1.321635 | 0.837563 | 1.105726 | 0.800833 | 0.952432 | 0.673960 | 1.327502 | 0.806885 | 1.135634 |      |      |      |      |      |
|       | 0.722275 | 1.482196 | 1.269134 | 0.695424 | 0.734410 | 1.240968 | 0.873750 | 0.825209 | 0.794290 | 0.840055 |      |      |      |      |      |
|       | 1.022015 | 1.408327 | 0.702067 | 0.966478 | 0.742264 | 1.204704 | 0.633885 | 0.888498 | 0.713166 | 1.231153 |      |      |      |      |      |
|       | 0.512817 | 1.187577 | 0.939995 | 1.273059 | 1.261853 | 0.875396 | 0.884657 | 0.928591 | 0.900721 | 0.880112 |      |      |      |      |      |
|       | 0.781537 | 1.223652 | 0.854824 | 0.972599 | 1.347144 | 0.807227 | 0.911694 | 0.956919 | 0.990262 | 0.893345 |      |      |      |      |      |
|       | 0.358832 | 1.601542 | 1.125781 | 0.730841 | 0.795475 | 0.856229 | 1.070817 | 0.869601 | 1.043598 | 0.896567 |      |      |      |      |      |
|       | 0.821456 | 1.676333 | 0.931807 | 0.740210 | 0.946762 | 0.917425 | 0.936539 | 0.884692 | 0.872158 | 0.827619 |      |      |      |      |      |
|       | 1.133434 | 1.069057 | 0.628708 | 1.348112 | 1.024242 | 0.726333 | 0.485013 | 0.517711 | 0.812878 | 1.483409 |      |      |      |      |      |
|       | 1.439495 | 0.401963 | 0.704352 | 1.185582 | 1.196889 | 0.894400 | 0.762308 | 0.871978 | 1.004571 | 0.965074 |      |      |      |      |      |

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TABLE VIII. Codebook used by the Fourier magnitude vector quantizer - Continued.

| Index | Vectors  |          |          |          |          |          |          |          |          |          |
|-------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| 0x62  | 1.355857 | 1.217795 | 0.623188 | 0.626000 | 1.080819 | 1.443039 | 0.690217 | 0.676470 | 0.752167 | 0.890824 |
| 0x63  | 1.384203 | 0.893545 | 0.571820 | 1.422045 | 1.054457 | 0.896980 | 0.929532 | 0.779836 | 0.816629 | 0.774798 |
| 0x64  | 1.272930 | 0.807536 | 0.943865 | 1.161822 | 1.109685 | 0.883353 | 0.756987 | 1.183318 | 0.792499 | 0.818768 |
| 0x65  | 1.368426 | 0.939970 | 0.927314 | 1.372501 | 0.966845 | 0.661820 | 0.817570 | 0.813094 | 0.874638 | 0.905751 |
| 0x66  | 1.234512 | 0.812394 | 1.092298 | 0.956999 | 0.926831 | 1.159350 | 0.560790 | 0.914404 | 1.104409 | 0.951357 |
| 0x67  | 1.462063 | 1.107453 | 0.736919 | 1.182041 | 0.753060 | 0.817381 | 0.845724 | 0.960321 | 0.930034 | 0.870101 |
| 0x68  | 1.154263 | 1.149050 | 0.969239 | 1.003195 | 0.709246 | 1.334899 | 0.539388 | 0.648414 | 1.170228 | 0.841587 |
| 0x69  | 1.363427 | 0.856319 | 0.848329 | 1.330223 | 1.580245 | 0.593904 | 0.534564 | 0.577523 | 0.647285 | 0.799871 |
| 0x6a  | 1.430748 | 1.277622 | 0.535910 | 1.316419 | 0.704018 | 1.139968 | 0.844730 | 0.708186 | 0.689915 | 0.673166 |
| 0x6b  | 1.456831 | 0.852769 | 0.836842 | 1.661979 | 0.555156 | 0.659789 | 0.754965 | 0.765223 | 0.801905 | 0.871901 |
| 0x6c  | 1.219340 | 0.941999 | 1.021315 | 1.258016 | 0.869648 | 1.086416 | 1.006363 | 0.806097 | 0.807889 | 0.739562 |
| 0x6d  | 1.269435 | 1.217910 | 1.091920 | 1.130622 | 1.241678 | 0.548636 | 0.659441 | 0.769661 | 0.738142 | 0.844145 |
| 0x6e  | 1.109102 | 1.109600 | 0.921873 | 1.080103 | 0.644065 | 1.551581 | 0.917328 | 0.782032 | 0.649409 | 0.682361 |
| 0x6f  | 1.148823 | 1.115256 | 0.784950 | 1.519787 | 0.632186 | 0.880947 | 0.965093 | 0.738516 | 0.923041 | 0.843738 |
| 0x70  | 1.099162 | 1.072237 | 1.059239 | 1.012868 | 1.091543 | 0.820111 | 0.632757 | 0.796845 | 1.140959 | 1.032531 |
| 0x71  | 1.382078 | 1.014650 | 0.855020 | 1.088445 | 1.147617 | 1.054602 | 0.759230 | 0.758359 | 0.828497 | 0.830315 |
| 0x72  | 1.331108 | 0.911897 | 0.551680 | 0.739304 | 1.080511 | 1.173246 | 0.671436 | 0.925538 | 1.082333 | 1.110124 |
| 0x73  | 1.380841 | 1.130295 | 0.743747 | 0.910074 | 0.957719 | 0.986217 | 0.885270 | 0.809240 | 0.914088 | 1.051811 |
| 0x74  | 1.310270 | 0.998818 | 0.989857 | 1.025202 | 0.906801 | 0.876459 | 0.895856 | 0.981348 | 0.982188 | 0.901957 |
| 0x75  | 1.442392 | 0.947518 | 1.052543 | 0.888925 | 1.092888 | 0.674498 | 0.920650 | 0.987177 | 0.890263 | 0.816537 |
| 0x76  | 1.173251 | 0.879037 | 1.128982 | 0.905439 | 1.302826 | 1.072854 | 0.709855 | 0.689286 | 1.000834 | 0.819403 |
| 0x77  | 1.162547 | 0.939193 | 1.038151 | 0.930689 | 0.914292 | 0.877584 | 1.144403 | 0.667857 | 1.257437 | 0.810249 |
| 0x78  | 0.957634 | 1.145630 | 1.008914 | 1.045813 | 1.030746 | 1.071616 | 0.801657 | 0.909400 | 0.880216 | 1.014015 |
| 0x79  | 1.250553 | 1.148173 | 0.606277 | 0.991795 | 0.997859 | 0.688890 | 0.942568 | 0.922857 | 1.233180 | 0.891771 |
| 0x7a  | 1.223899 | 1.343365 | 0.813470 | 0.600631 | 0.514261 | 1.292102 | 0.994158 | 1.012705 | 0.774822 | 0.889761 |

TABLE VIII. Codebook used by the Fourier magnitude vector quantizer - Continued.

| Index | Vectors  |          |          |          |          |          |          |          |          |          |
|-------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| 0x7b  | 1.434258 | 1.190874 | 0.478282 | 0.454840 | 0.731792 | 1.170314 | 0.776934 | 1.201793 | 0.865726 | 0.993823 |
| 0x7c  | 1.075483 | 0.950940 | 1.056806 | 0.939311 | 0.767863 | 1.050672 | 0.923138 | 1.208091 | 0.957328 | 0.914622 |
| 0x7d  | 1.152850 | 0.985329 | 0.758860 | 0.902903 | 1.268339 | 0.822622 | 0.951593 | 0.972154 | 0.988798 | 1.010038 |
| 0x7e  | 0.934316 | 1.213602 | 1.137776 | 0.530840 | 0.583216 | 1.327735 | 0.648220 | 1.122533 | 0.983140 | 0.947688 |
| 0x7f  | 1.096461 | 1.329950 | 1.056696 | 1.096312 | 0.518753 | 0.667957 | 1.377914 | 0.802439 | 0.713627 | 0.728387 |
| 0x80  | 0.438479 | 0.986558 | 0.795180 | 1.048475 | 1.212747 | 1.336801 | 0.859633 | 1.040980 | 0.943136 | 0.921149 |
| 0x81  | 1.106514 | 0.482598 | 0.612108 | 1.642264 | 0.798236 | 0.780622 | 0.825869 | 0.883680 | 1.122055 | 0.975218 |
| 0x82  | 0.977340 | 0.929082 | 0.748423 | 0.688517 | 1.866792 | 0.978102 | 0.704420 | 0.691328 | 0.814419 | 0.732273 |
| 0x83  | 2.061007 | 1.317233 | 0.531080 | 0.663650 | 0.466228 | 0.673965 | 0.663756 | 0.680624 | 0.722060 | 0.735546 |
| 0x84  | 0.324483 | 0.566419 | 1.626532 | 0.909957 | 0.767043 | 1.218518 | 0.923857 | 0.781191 | 1.034489 | 0.929690 |
| 0x85  | 1.132776 | 0.664024 | 0.658276 | 1.115888 | 1.058611 | 0.773592 | 1.143017 | 1.345343 | 0.885859 | 0.793138 |
| 0x86  | 0.693193 | 0.957946 | 1.074530 | 0.991271 | 1.178869 | 1.050262 | 1.231718 | 0.919607 | 0.838514 | 0.820971 |
| 0x87  | 0.981206 | 1.282808 | 0.628439 | 0.831905 | 0.871888 | 1.000360 | 1.072068 | 1.359583 | 0.865264 | 0.714229 |
| 0x88  | 0.444708 | 1.320901 | 0.863009 | 0.823633 | 1.282252 | 1.175766 | 1.016159 | 0.800445 | 0.885734 | 0.918392 |
| 0x89  | 0.524278 | 0.903183 | 0.842282 | 0.999358 | 1.663203 | 0.862494 | 0.990065 | 0.863742 | 0.861246 | 0.885159 |
| 0x8a  | 0.647517 | 1.630489 | 0.674425 | 0.552789 | 1.367019 | 0.802523 | 1.078554 | 0.810538 | 0.843750 | 0.824923 |
| 0x8b  | 1.157635 | 1.168651 | 0.661005 | 0.556302 | 1.240422 | 0.970912 | 0.829631 | 1.237884 | 0.917322 | 0.831318 |
| 0x8c  | 0.271398 | 0.978008 | 0.699287 | 1.501347 | 0.903792 | 1.105029 | 0.960131 | 1.036695 | 1.001109 | 0.890359 |
| 0x8d  | 0.275143 | 0.628587 | 1.143603 | 1.112293 | 1.315251 | 1.071179 | 0.889841 | 0.954181 | 1.018312 | 0.964080 |
| 0x8e  | 0.659209 | 1.447333 | 0.894087 | 1.001992 | 0.839399 | 1.449942 | 0.796803 | 0.843993 | 0.756016 | 0.817636 |
| 0x8f  | 0.853138 | 1.064738 | 0.855133 | 0.742170 | 0.868512 | 1.370756 | 0.989681 | 1.038327 | 0.937196 | 0.993200 |
| 0x90  | 0.973539 | 1.188932 | 1.023116 | 0.931037 | 1.059522 | 0.699512 | 0.709015 | 1.360161 | 0.889684 | 0.811761 |
| 0x91  | 1.226588 | 1.022695 | 0.719215 | 1.364611 | 0.676914 | 0.710196 | 1.255418 | 1.207082 | 0.643748 | 0.569250 |
| 0x92  | 0.911359 | 1.344114 | 1.042414 | 0.518823 | 1.545711 | 0.394875 | 0.583463 | 1.042424 | 0.973970 | 0.781663 |
| 0x93  | 1.728382 | 0.630457 | 0.384633 | 0.410263 | 0.623340 | 0.756137 | 0.963416 | 1.240436 | 1.179740 | 0.957703 |

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TABLE VIII. Codebook used by the Fourier magnitude vector quantizer - Continued.

| Index | Vectors  |          |          |          |          |          |          |          |          |          |  |  |  |  |  |
|-------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|--|--|--|--|--|
|       |          |          |          |          |          |          |          |          |          |          |  |  |  |  |  |
| 0x94  | 0.738959 | 0.590931 | 1.110580 | 1.101743 | 0.983178 | 0.969963 | 1.084357 | 1.136055 | 1.021410 | 0.944931 |  |  |  |  |  |
| 0x95  | 1.244427 | 0.569958 | 1.057313 | 1.369461 | 0.723367 | 0.805159 | 1.027868 | 1.091877 | 0.873967 | 0.776658 |  |  |  |  |  |
| 0x96  | 0.787743 | 0.905228 | 1.203273 | 0.843008 | 1.182612 | 1.029414 | 0.763719 | 1.097128 | 0.900016 | 1.016990 |  |  |  |  |  |
| 0x97  | 1.314427 | 1.012982 | 0.695385 | 0.878737 | 0.749022 | 0.792771 | 1.048592 | 1.421864 | 0.816344 | 0.837093 |  |  |  |  |  |
| 0x98  | 0.284264 | 1.037450 | 0.783668 | 1.045137 | 1.085681 | 0.925955 | 1.010468 | 0.963587 | 1.285465 | 1.083612 |  |  |  |  |  |
| 0x99  | 0.182413 | 0.501944 | 0.613194 | 0.864945 | 1.046530 | 1.201125 | 1.222115 | 1.201816 | 1.144754 | 1.100226 |  |  |  |  |  |
| 0x9a  | 0.399720 | 1.302627 | 1.339572 | 0.544091 | 1.113712 | 0.704227 | 0.959569 | 1.086265 | 0.919326 | 0.996120 |  |  |  |  |  |
| 0x9b  | 1.312087 | 1.234779 | 1.155127 | 0.573204 | 0.603170 | 1.675500 | 0.578462 | 0.549438 | 0.586763 | 0.627105 |  |  |  |  |  |
| 0x9c  | 0.347498 | 1.019206 | 0.902255 | 0.918363 | 1.078096 | 0.895444 | 1.295650 | 1.236476 | 0.896594 | 0.955950 |  |  |  |  |  |
| 0x9d  | 0.228297 | 0.547356 | 1.316811 | 1.334579 | 0.850478 | 0.739122 | 1.162684 | 1.153175 | 0.961239 | 0.847730 |  |  |  |  |  |
| 0x9e  | 0.450951 | 1.164991 | 1.356497 | 0.748838 | 1.098313 | 1.180630 | 0.960353 | 0.882357 | 0.871493 | 0.801907 |  |  |  |  |  |
| 0x9f  | 1.101005 | 1.177952 | 1.577841 | 0.463778 | 0.535020 | 0.573959 | 0.857461 | 1.214102 | 0.824723 | 0.827632 |  |  |  |  |  |
| 0xa0  | 1.008192 | 0.512187 | 0.511630 | 0.803656 | 1.363784 | 1.397129 | 1.041883 | 0.945399 | 0.880500 | 0.836761 |  |  |  |  |  |
| 0xa1  | 1.333032 | 0.522169 | 0.512363 | 0.626419 | 0.780427 | 0.951713 | 1.495366 | 1.235918 | 0.888195 | 0.836141 |  |  |  |  |  |
| 0xa2  | 1.680621 | 0.528954 | 0.445504 | 0.623836 | 1.316133 | 1.053039 | 0.890863 | 0.877480 | 0.775161 | 0.883213 |  |  |  |  |  |
| 0xa3  | 2.235511 | 0.589895 | 0.429925 | 0.577054 | 0.680209 | 0.667395 | 0.784315 | 0.803936 | 0.861809 | 0.801209 |  |  |  |  |  |
| 0xa4  | 1.092796 | 0.555938 | 1.481165 | 0.731586 | 1.165349 | 0.972450 | 0.847847 | 0.832928 | 0.893096 | 0.925555 |  |  |  |  |  |
| 0xa5  | 1.276456 | 0.719658 | 0.981702 | 1.037082 | 1.176864 | 0.684247 | 0.782879 | 0.891723 | 1.117659 | 1.034450 |  |  |  |  |  |
| 0xa6  | 1.228245 | 0.771626 | 1.121788 | 0.749934 | 0.884593 | 0.958108 | 1.233677 | 1.045060 | 0.933820 | 0.833295 |  |  |  |  |  |
| 0xa7  | 1.469187 | 0.712424 | 0.977325 | 0.707326 | 1.237597 | 0.953937 | 0.835168 | 0.794138 | 0.991406 | 0.934207 |  |  |  |  |  |
| 0xa8  | 1.079164 | 1.065143 | 1.263724 | 0.575489 | 1.628970 | 0.516204 | 0.546497 | 0.527023 | 0.774807 | 1.079119 |  |  |  |  |  |
| 0xa9  | 1.337945 | 0.994695 | 0.675169 | 0.865744 | 1.373516 | 1.324911 | 0.866779 | 0.633895 | 0.599425 | 0.727921 |  |  |  |  |  |
| 0xaa  | 1.511212 | 0.641169 | 1.710560 | 0.731522 | 0.590315 | 0.546117 | 0.624898 | 0.778970 | 0.908906 | 0.903649 |  |  |  |  |  |
| 0xab  | 1.760845 | 1.236091 | 1.067203 | 0.487204 | 0.655720 | 0.713362 | 0.697764 | 0.749475 | 0.847174 | 0.921298 |  |  |  |  |  |
| 0xac  | 1.242002 | 1.065400 | 0.986706 | 0.626575 | 1.306310 | 0.681929 | 0.717473 | 0.615010 | 1.350818 | 0.832838 |  |  |  |  |  |

| Index | Vectors  |          |          |          |          |          |          |          |          |          |  |  |  |  |  |
|-------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|--|--|--|--|--|
| 0xad  | 1.114715 | 0.651334 | 1.082160 | 1.095767 | 1.064034 | 1.122643 | 1.004475 | 0.903535 | 0.874168 | 0.865362 |  |  |  |  |  |
| 0xae  | 1.244497 | 1.005714 | 1.193496 | 0.734583 | 0.598761 | 1.274922 | 1.289539 | 0.614727 | 0.688696 | 0.778442 |  |  |  |  |  |
| 0xaf  | 1.408250 | 0.839767 | 1.067828 | 0.572007 | 0.672582 | 1.239141 | 0.865001 | 0.858314 | 1.080318 | 0.941161 |  |  |  |  |  |
| 0xb0  | 1.250488 | 0.918235 | 1.101326 | 0.668242 | 1.365644 | 0.555295 | 0.552784 | 1.259466 | 0.907460 | 0.811412 |  |  |  |  |  |
| 0xb1  | 1.631045 | 0.643427 | 0.448596 | 0.462433 | 0.721966 | 1.376064 | 1.149863 | 0.918544 | 0.813496 | 0.874574 |  |  |  |  |  |
| 0xb2  | 1.535075 | 1.109493 | 1.002904 | 0.867410 | 1.515072 | 0.754291 | 0.569348 | 0.512821 | 0.594932 | 0.664862 |  |  |  |  |  |
| 0xb3  | 1.968050 | 0.717136 | 1.151858 | 0.741070 | 0.764881 | 0.686026 | 0.715694 | 0.699206 | 0.765000 | 0.778737 |  |  |  |  |  |
| 0xb4  | 1.413788 | 0.795490 | 1.377022 | 0.733397 | 1.230761 | 0.728025 | 0.815985 | 0.866154 | 0.713555 | 0.821627 |  |  |  |  |  |
| 0xb5  | 1.339077 | 0.710095 | 1.014431 | 1.092420 | 0.988468 | 0.688934 | 1.213969 | 0.865531 | 0.928889 | 0.852068 |  |  |  |  |  |
| 0xb6  | 1.172569 | 0.971105 | 1.147345 | 0.503528 | 1.077878 | 1.006880 | 0.936191 | 1.024685 | 0.978125 | 0.897520 |  |  |  |  |  |
| 0xb7  | 1.346133 | 0.825702 | 1.278602 | 0.712776 | 0.990801 | 0.824307 | 0.741063 | 0.777009 | 1.172757 | 0.958583 |  |  |  |  |  |
| 0xb8  | 0.838360 | 1.061588 | 1.184451 | 0.806865 | 1.445930 | 0.716610 | 1.122729 | 0.763437 | 0.824729 | 0.830950 |  |  |  |  |  |
| 0xb9  | 1.276108 | 0.881797 | 1.013445 | 0.821855 | 1.264900 | 1.025945 | 1.227647 | 0.703569 | 0.659312 | 0.734361 |  |  |  |  |  |
| 0xba  | 1.043635 | 1.011762 | 1.513310 | 0.739962 | 0.660585 | 0.587323 | 0.680578 | 0.841721 | 1.167268 | 1.132996 |  |  |  |  |  |
| 0xbb  | 1.473979 | 0.627258 | 1.381177 | 0.659284 | 0.759202 | 1.049849 | 1.020305 | 0.870139 | 0.779335 | 0.834162 |  |  |  |  |  |
| 0xbc  | 1.308547 | 0.926336 | 1.119570 | 0.550398 | 1.232051 | 0.605843 | 1.181402 | 0.819554 | 0.865953 | 0.920083 |  |  |  |  |  |
| 0xbd  | 1.337645 | 0.769449 | 1.229177 | 0.813618 | 0.735630 | 0.883470 | 1.569295 | 0.634285 | 0.606052 | 0.688148 |  |  |  |  |  |
| 0xbe  | 1.290794 | 1.093894 | 1.145434 | 0.552063 | 0.804350 | 0.769436 | 0.840595 | 1.089482 | 1.009731 | 1.051359 |  |  |  |  |  |
| 0xbf  | 1.364932 | 0.913519 | 1.316632 | 0.718362 | 0.529225 | 0.723872 | 1.181281 | 0.876150 | 0.962565 | 0.918262 |  |  |  |  |  |
| 0xc0  | 0.803741 | 1.388152 | 0.980932 | 1.386743 | 0.402886 | 1.229764 | 0.855613 | 0.743859 | 0.738745 | 0.812385 |  |  |  |  |  |
| 0xc1  | 1.187083 | 1.328233 | 0.750736 | 1.808456 | 0.498683 | 0.428506 | 0.498878 | 0.815022 | 0.756569 | 0.712230 |  |  |  |  |  |
| 0xc2  | 1.008780 | 1.634667 | 0.998899 | 1.114864 | 1.106568 | 0.814867 | 0.677242 | 0.626767 | 0.727848 | 0.701162 |  |  |  |  |  |
| 0xc3  | 1.442123 | 1.578969 | 1.235055 | 1.279799 | 0.604508 | 0.459123 | 0.486663 | 0.548079 | 0.537477 | 0.639041 |  |  |  |  |  |
| 0xc4  | 0.824699 | 0.832512 | 1.286809 | 0.946005 | 0.788899 | 1.581450 | 0.819352 | 0.804348 | 0.757910 | 0.797158 |  |  |  |  |  |
| 0xc5  | 1.173084 | 0.852089 | 0.837159 | 1.215371 | 0.614107 | 0.664921 | 0.798941 | 1.185092 | 1.404650 | 0.752108 |  |  |  |  |  |

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TABLE VIII. Codebook used by the Fourier magnitude vector quantizer - Continued.

| Index | Vectors  |          |          |          |          |          |          |          |          |  |  |  |  |          |  |
|-------|----------|----------|----------|----------|----------|----------|----------|----------|----------|--|--|--|--|----------|--|
|       |          |          |          |          |          |          |          |          |          |  |  |  |  |          |  |
| 0xc6  | 0.837980 | 1.195548 | 1.251800 | 1.166862 | 1.139151 | 0.630036 | 0.966812 | 0.823920 | 0.819578 |  |  |  |  | 0.833867 |  |
| 0xc7  | 1.284059 | 1.360185 | 0.903226 | 1.260066 | 0.869280 | 0.794052 | 0.767674 | 0.784999 | 0.770146 |  |  |  |  | 0.814413 |  |
| 0xc8  | 0.434909 | 1.062061 | 0.820721 | 0.980006 | 0.915903 | 1.352711 | 1.363232 | 0.818512 | 0.933755 |  |  |  |  | 0.804752 |  |
| 0xc9  | 0.140705 | 0.369673 | 0.761460 | 1.721146 | 1.281940 | 0.814390 | 0.770430 | 0.912685 | 1.029093 |  |  |  |  | 0.894100 |  |
| 0xca  | 0.381833 | 1.386068 | 0.895360 | 0.756433 | 0.795777 | 0.665474 | 1.098763 | 1.438104 | 1.037485 |  |  |  |  | 0.828884 |  |
| 0xcb  | 0.848084 | 1.088554 | 0.824684 | 0.776900 | 0.721938 | 0.676660 | 0.800316 | 1.135752 | 1.511374 |  |  |  |  | 1.057070 |  |
| 0xcc  | 0.675218 | 1.083363 | 1.015844 | 1.396638 | 0.903676 | 0.983177 | 0.888414 | 1.062584 | 0.890620 |  |  |  |  | 0.821218 |  |
| 0xcd  | 0.754970 | 0.952752 | 1.127383 | 1.539338 | 0.700672 | 0.751028 | 0.640102 | 0.797817 | 1.188132 |  |  |  |  | 0.971816 |  |
| 0xce  | 0.701999 | 1.321737 | 1.118923 | 1.317673 | 0.798997 | 1.146361 | 0.653062 | 0.923909 | 0.716226 |  |  |  |  | 0.881342 |  |
| 0xcf  | 0.916108 | 1.387334 | 0.769658 | 1.200061 | 0.617306 | 1.096865 | 0.828575 | 1.077143 | 0.950269 |  |  |  |  | 0.767253 |  |
| 0xd0  | 0.662335 | 1.292372 | 1.294051 | 1.134329 | 0.611097 | 0.878492 | 0.979876 | 0.932184 | 0.972586 |  |  |  |  | 0.870668 |  |
| 0xd1  | 1.081800 | 1.056505 | 1.431825 | 1.568379 | 0.608668 | 0.601313 | 0.533137 | 0.646402 | 0.768233 |  |  |  |  | 0.839110 |  |
| 0xd2  | 1.062379 | 1.439992 | 1.195124 | 1.101291 | 0.690349 | 0.686199 | 0.744970 | 0.782839 | 0.947571 |  |  |  |  | 0.913290 |  |
| 0xd3  | 1.302173 | 1.079498 | 1.059178 | 0.961715 | 0.721412 | 0.548559 | 0.490001 | 0.649919 | 1.527619 |  |  |  |  | 0.917298 |  |
| 0xd4  | 0.680923 | 1.193324 | 1.324460 | 1.076160 | 0.940757 | 1.158478 | 1.016504 | 0.799201 | 0.766968 |  |  |  |  | 0.708056 |  |
| 0xd5  | 1.013158 | 0.953016 | 1.286560 | 1.218908 | 0.634730 | 1.131540 | 0.788277 | 0.928104 | 0.871793 |  |  |  |  | 0.842399 |  |
| 0xd6  | 0.801060 | 0.948640 | 1.501946 | 0.737009 | 0.751623 | 1.024697 | 1.091762 | 0.879438 | 0.970670 |  |  |  |  | 0.874395 |  |
| 0xd7  | 1.322228 | 1.272781 | 1.083638 | 0.879445 | 0.649796 | 0.924154 | 0.851784 | 0.949572 | 0.890756 |  |  |  |  | 0.876190 |  |
| 0xd8  | 0.308295 | 1.165975 | 1.273902 | 1.153456 | 0.938787 | 0.761951 | 0.988352 | 0.937937 | 0.984994 |  |  |  |  | 0.988803 |  |
| 0xd9  | 0.307028 | 0.833986 | 1.891761 | 1.166166 | 0.526468 | 0.545358 | 0.772629 | 0.838092 | 0.849941 |  |  |  |  | 1.006412 |  |
| 0xda  | 0.724400 | 1.661022 | 1.407543 | 0.659395 | 0.621709 | 0.664584 | 0.781703 | 0.915312 | 0.912252 |  |  |  |  | 0.868291 |  |
| 0xdb  | 0.899730 | 0.520003 | 2.050203 | 0.570193 | 0.525624 | 0.618870 | 0.773975 | 0.953262 | 0.860057 |  |  |  |  | 0.852288 |  |
| 0xdc  | 0.564036 | 1.194072 | 1.054038 | 1.309234 | 0.682175 | 1.184083 | 1.261020 | 0.720925 | 0.837966 |  |  |  |  | 0.682236 |  |
| 0xdd  | 0.705321 | 0.906931 | 1.440192 | 1.246568 | 0.945792 | 0.911725 | 0.914322 | 0.903469 | 0.819844 |  |  |  |  | 0.853160 |  |
| 0xde  | 0.639259 | 1.118893 | 1.183435 | 0.912825 | 0.722814 | 1.107206 | 0.903505 | 1.289695 | 0.937470 |  |  |  |  | 0.852877 |  |

## Vectors

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TABLE VIII. Codebook used by the Fourier magnitude vector quantizer - Continued.

| Index | Vectors  |          |          |          |          |          |          |          |          |          |
|-------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
|       | 1.087334 | 1.237373 | 1.295267 | 0.657919 | 1.094758 | 1.035074 | 0.693941 | 0.869327 | 0.868297 | 0.787626 |
| 0xf8  | 1.183917 | 0.792976 | 1.143049 | 0.749982 | 0.626687 | 0.658678 | 0.983599 | 1.416025 | 1.044253 | 0.931234 |
| 0xf9  | 1.694116 | 1.108393 | 1.504700 | 0.828677 | 0.586589 | 0.658925 | 0.648047 | 0.676080 | 0.628926 | 0.681370 |
| 0xfa  | 1.363659 | 0.859205 | 1.134321 | 1.227826 | 0.519545 | 0.570781 | 0.847817 | 1.035284 | 0.967904 | 0.993060 |
| 0xfb  | 1.202407 | 0.995001 | 1.034142 | 0.944991 | 0.778536 | 0.657893 | 1.164819 | 0.964661 | 0.913810 | 1.114029 |
| 0xfc  | 1.325400 | 0.953743 | 1.087893 | 1.003095 | 0.754906 | 0.862034 | 1.119859 | 1.201467 | 0.736865 | 0.616466 |
| 0xfe  | 1.321790 | 0.767809 | 1.313064 | 0.954328 | 0.816200 | 0.607967 | 0.663671 | 1.355803 | 0.816644 | 0.850276 |
| 0xff  | 1.493994 | 1.117090 | 1.211209 | 1.311853 | 0.619853 | 0.708056 | 0.764515 | 0.708914 | 0.718302 | 0.743372 |

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## MELP ALGORITHM DESCRIPTION

## A.1 SCOPE

A.1.1 Scope. This appendix provides a complete description of a MELP algorithm. This appendix is not a mandatory part of this standard. The information contained herein is intended for guidance only.

## A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents. The documents in 2. of this standard apply to this appendix.

A.2.2 Other publications. The following documents form a part of this appendix to the extent specified.

## INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

"A Mixed Excitation LPC Vocoder Model for Low Bit Rate Speech Coding"  
by A.V. McCree and T.P. Barnwell III, Transactions on Speech and Audio  
Processing, Vol. 3, No. 4, July 1995, 242-250

"A 2.4 kbits/s MELP Coder Candidate for the New U.S. Federal Standard"  
by A. McCree, K. Truong, E.B. George, T.P. Barnwell III, and V. Viswanathan,  
Proceedings of IEEE ICASSP 1996, pp.200-203

"Super Resolution Pitch Determination of Speech Signals"  
by Y. Medan, E. Yair, and D. Chazan, Transactions on Signal Processing, Vol. 39,  
No. 1, January 1991, pp. 40-48

"The Computation of Line Spectral Frequencies Using Chebyshev Polynomials"  
by P. Kabal and R.P. Ramachandran, Transactions on Acoustics, Speech, and Signal  
Processing, Vol. ASSP-34, No.6, December 1986, pp.1419-1426

"Efficient Search and Design Procedures for Robust Multi-Stage VQ of LPC  
Parameters for 4 kb/s Speech Coding"  
by W.P. LeBlanc, B. Bhattacharya, S.A. Mahmoud, and V. Cuperman, Transactions  
on Speech and Audio Processing, Vol. 1, No. 4, October 1993, pp. 373-385

"New Methods for Adaptive Noise Suppression"  
by L. Arslan, A. McCree, and V. Viswanathan, Proceedings of IEEE ICASSP 1995,  
pp. 812-815

"MELP: The New Federal Standard at 2400 BPS"  
by L. Supplee, R. Cohn, J. Collura, A. McCree, Proceedings of IEEE ICASSP 1997,  
pp. 1591-1594

(Applications for copies should be addressed to IEEE Customer Service, 445 Hoes Lane, P.O. Box 1331  
Piscataway, New Jersey 08855-1331, USA)

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A.2.3 Order of precedence. In the event of a conflict between the text of this standard and the references stated herein, the text of this standard should take precedence.

## A.3 DEFINITIONS

A.3.1 Terms. The definitions in 3.1 of this standard apply to this appendix.

A.3.2 Acronyms. The acronyms used in this appendix are defined in 3.2 or as follows:

DC - Direct Current

DFT - Discrete Fourier Transform

FFT - Fast Fourier Transform

FIR - Finite Impulse Response

RMS - Root Mean Square

VQ - Vector Quantization

## A.4 GENERAL REQUIREMENTS

Not applicable.

## A.5 DETAILED REQUIREMENTS

A.5.1 General. The Mixed Excitation Linear Prediction coder is based on the traditional Linear Prediction Coding (LPC) parametric model, but also includes five additional features. These are: mixed excitation, aperiodic pulses, adaptive spectral enhancement, pulse dispersion, and Fourier magnitude modeling. These features are illustrated in the MELP decoder block diagram shown in Figure A-1.

The mixed excitation is implemented using a multi-band mixing model. This model can simulate frequency-dependent voicing strength using an adaptive filtering structure implemented with a fixed filter bank. The primary effect of this mixed excitation is to reduce the buzz usually associated with LPC vocoders, especially in broadband acoustic noise.

When the input speech is voiced, the MELP coder can synthesize using either periodic or aperiodic pulses. Aperiodic pulses are used most often during transition regions between voiced and unvoiced segments of the speech signal. This feature enables the decoder to reproduce erratic glottal pulses without introducing tonal sounds.

The adaptive spectral enhancement filter is based on the poles of the linear prediction synthesis filter. Its use enhances the formant structure of the synthetic speech and improves the match between the synthetic and natural bandpass waveforms. It also gives the synthetic speech a more natural quality.



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A buffer containing the most recent samples of the input speech signal is maintained in the encoder. One of these samples is designated the last sample in the current frame. The buffer extends beyond this sample into the past and future to contain the samples needed for the encoding process. The last sample in the current frame serves as a reference point for many of the encoder calculations.

**A5.2.2 Integer pitch calculation.** For this pitch calculation, the input speech signal is first processed with a 1 kHz, 6<sup>th</sup> order Butterworth lowpass filter. The integer pitch value,  $P_1$ , is the value of  $\tau$ ,  $\tau = 40, 41, \dots, 160$ , for which the normalized autocorrelation function,  $r(\tau)$ , is maximized. This function is defined by:

$$r(\tau) = \frac{c_\tau(0, \tau)}{\sqrt{c_\tau(0, 0)c_\tau(\tau, \tau)}}, \text{ EQUATION A-1,}$$

$$\text{where } c_\tau(m, n) = \sum_{k=-\lfloor \tau/2 \rfloor - 80}^{-\lfloor \tau/2 \rfloor + 79} s_{k+m} s_{k+n}, \text{ EQUATION A-2,}$$

and  $\lfloor \tau/2 \rfloor$  represents truncation to an integer value. The center of the pitch analysis window is at sample  $s_0$  in equation A-2. For the integer pitch calculation, this window is centered on the last sample in the current frame. The lowpass filter output is sample  $s_0$  when its input is the last sample in the current frame. The time index  $k$  in the autocorrelation preserves the pitch analysis window alignment around its center point; the normalization compensates for changing signal amplitudes. The final pitch calculation (see A5.2.9) extends the pitch range to a lag of 20 samples.

**A5.2.3 Bandpass voicing analysis.** This portion of the encoder determines the five bandpass voicing strengths,  $Vbp_i, i = 1, 2, \dots, 5$ . It also refines the integer pitch measurement and the corresponding normalized autocorrelation value. The bandpass voicing analysis begins by filtering the input speech signal into five frequency bands. These filters are 6<sup>th</sup> order Butterworth, with passbands of 0-500, 500-1000, 1000-2000, 2000-3000, and 3000-4000 Hz.

A refined pitch measurement is made using the 0-500 Hz filter output signal. This measurement is centered on the filter output produced when its input is the last sample in the current frame. Two pitch candidates are considered in this refinement, namely the integer pitch values,  $P_1$ , from the current and previous frames. For each candidate, equation A-1 is used to perform an integer pitch search over lags from 5 samples shorter to 5 samples longer than the candidate, and a fractional pitch refinement (see A5.2.4) is performed around the optimum integer pitch lag. This produces two fractional pitch candidates and their corresponding normalized autocorrelation values. The candidate having the higher normalized autocorrelation is selected as the fractional pitch,  $P_2$ . The corresponding normalized autocorrelation,  $r(P_2)$ , is saved as the lowest band voicing strength,  $Vbp_1$ .  $P_2$  is saved for use in determining the voicing strength for the remaining frequency bands. It is also used in the final pitch calculation (see A5.2.9) and gain calculation (see A5.2.11).

For each remaining band, the bandpass voicing strength is the larger of  $r(P_2)$  as determined by the fractional pitch procedure for the bandpass signal and the time envelope of the bandpass signal, where  $r(P_2)$  for the time envelope is first decremented by 0.1 to compensate for an experimentally observed bias (due to the smoothness of the time envelope signals). The envelopes are calculated by full-wave rectification followed by a smoothing filter. This filter consists of a zero at DC in cascade with a complex pole pair at 150 Hz with a radius of 0.97. For each

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calculation of  $r(p_2)$ , the analysis window is centered on the last sample in the current frame, as was the case for the first band.

**A5.2.4 Fractional pitch refinement.** This procedure, which is used at several places in the encoding process, utilizes an interpolation formula to increase the accuracy of an input pitch value. This value is first rounded to the nearest integer. Assume that this integer has a value of  $T$  samples. The interpolation formula presumes that  $r(\tau)$  has a maximum between lags of  $T$  and  $T+1$ . Hence,  $c_T(0, T-1)$  and  $c_T(0, T+1)$  are computed and compared to determine if the maximum is more likely to fall between  $T$  and  $T+1$  or between  $T-1$  and  $T$ . If  $c_T(0, T-1) > c_T(0, T+1)$ , then the maximum probably falls between  $T-1$  and  $T$  and the pitch,  $T$ , is decremented by one prior to interpolation. The fractional offset,  $\Delta$ , is then computed by the interpolation equation:

$$\Delta = \frac{c_T(0, T+1)c_T(T, T) - c_T(0, T)c_T(T, T+1)}{c_T(0, T+1)[c_T(T, T) - c_T(T, T+1)] + c_T(0, T)[c_T(T+1, T+1) - c_T(T, T+1)]}, \text{ EQUATION A-3,}$$

where  $c_T(m, n)$  is defined by equation A-2. In some cases, this formula produces an offset outside the range of 0.0 to 1.0, so the offset is clamped between -1 and 2. The fractional pitch is  $T + \Delta$  and is clamped between 20 and 160.

The normalized autocorrelation at the fractional pitch value is given by:

$$r(T + \Delta) = \frac{(1 - \Delta)c_T(0, T) + \Delta c_T(0, T + 1)}{\sqrt{c_T(0, 0)[(1 - \Delta)^2 c_T(T, T) + 2\Delta(1 - \Delta)c_T(T, T + 1) + \Delta^2 c_T(T + 1, T + 1)]}}, \text{ EQUATION A-4.}$$

Equations A-3 and A-4 produce the fractional offset and corresponding normalized autocorrelation which would be obtained if the input signal had been linearly interpolated to obtain values between the actual sampling times.

**A5.2.5 Aperiodic flag.** The aperiodic flag is set to 1 if  $V_{bp1} < 0.5$  and set to 0 otherwise. The  $V_{bp1}$  value determined by bandpass voicing analysis (see A5.2.3) is used for this comparison. When set, this flag tells the decoder that the pulse component of the excitation should be aperiodic, rather than periodic. A5.3.1 describes the use of the aperiodic flag.

**A5.2.6 Linear prediction analysis.** A  $10^{\text{th}}$  order linear prediction analysis is performed on the input speech signal using a 200 sample (25 ms) Hamming window centered on the last sample in the current frame. The traditional autocorrelation analysis procedure is implemented using the Levinson-Durbin recursion. In addition, a bandwidth expansion coefficient of 0.994 (15 Hz) is applied to the prediction coefficients,  $a_i$ ,  $i = 1, 2, \dots, 10$ , where each coefficient is multiplied by  $0.994^i$ .

**A5.2.7 Linear prediction residual calculation.** The linear prediction residual signal is calculated by filtering the input speech signal with the prediction filter whose coefficients were determined by linear prediction analysis (see A5.2.6). The residual window is centered on the last sample in the current frame, and is made wide enough for use by the final pitch calculation (see A5.2.9).

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A5.2.8 Peakiness calculation. The peakiness of the residual signal is calculated over a 160 sample window centered on the last sample in the current frame. The peakiness value is the ratio of the L2 norm to the L1 norm of the residual signal,  $r_n$ , in the window:

$$\text{peakiness} = \frac{\sqrt{\frac{1}{160} \sum_{n=1}^{160} r_n^2}}{\frac{1}{160} \sum_{n=1}^{160} |r_n|}, \text{ EQUATION A-5.}$$

If the peakiness exceeds 1.34, then the lowest band voicing strength,  $Vbp_1$ , is forced to 1.0. If the peakiness exceeds 1.6, then the lowest three band voicing strengths,  $Vbp_i, i = 1, 2, 3$ , are all forced to 1.0. This is the only use of the peakiness measure.

A5.2.9 Final pitch calculation. The final pitch measurement uses the lowpass filtered residual signal, where the filter is a 6<sup>th</sup> order Butterworth, with a 1 kHz cutoff. Equation A-1 is used to perform an integer pitch search over lags from 5 samples shorter to 5 samples longer than  $P_2$ , rounded to the nearest integer. This measurement is centered on the filter output produced when its input is the last residual sample in the current frame. A fractional pitch refinement (see A5.2.4) is then made around the optimum integer pitch lag. This produces tentative values for the final pitch,  $P_3$ , and for the corresponding normalized autocorrelation,  $r(P_3)$ .

If  $r(P_3) \geq 0.6$ , the pitch doubling check procedure (see A5.2.10) is performed on the filtered residual, using  $P_3$  as the candidate pitch, and doubling threshold  $D_{th} = 0.75$  if  $P_3 \leq 100$ , or  $D_{th} = 0.5$  otherwise. The doubling check procedure may produce new values for  $P_3$  and  $r(P_3)$ .

The else action for the preceding if is as follows. A fractional pitch refinement around  $P_2$  is performed using the input speech signal. This measurement is centered on the last sample in the current frame and produces new values for  $P_3$  and  $r(P_3)$ . If  $r(P_3) < 0.55$ , then  $P_3$  is replaced by  $P_{avg}$ , the long-term average pitch (see A5.2.12). Otherwise, the pitch doubling check procedure is performed on the input speech signal, using  $P_3$  as the candidate pitch, and doubling threshold  $D_{th} = 0.9$  if  $P_3 \leq 100$ , or  $D_{th} = 0.7$  otherwise. The doubling check procedure may produce new values for  $P_3$  and  $r(P_3)$ .

Finally, if  $r(P_3) < 0.55$ , then  $P_3$  is replaced by  $P_{avg}$ .

The following pseudo code shows the final pitch algorithm:

inputs: the input speech signal; the residual signal;  $P_2$ ;  $P_{avg}$   
outputs:  $P_3$ ,  $cor\_P3$

```
fresid buffer = filter the residual with a 1 kHz Butterworth
P3 = best integer pitch on fresid over the range P2-5 to P2+5
P3, cor_P3 = frac_pitch(fresid, P3)
if (cor_P3 >= 0.6)
```

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```

    Dth = 0.5
    if (P3 <= 100) Dth = 0.75
    P3, cor_P3 = double_ck(fresid, P3, Dth)
else
    P3, cor_P3 = frac_pitch(input, P2)
    if (cor_P3 < 0.55)
        P3 = Pavg
    else
        Dth = 0.7
        if (P3 <= 100) Dth = 0.9
        P3, cor_P3 = double_ck(input, P3, Dth)
    endif
endif
if (cor_P3 < 0.55) P3 = Pavg

```

A5.2.10 Pitch doubling check. The pitch doubling check procedure looks for and corrects pitch values which are multiples of the actual pitch. This procedure takes a signal, a candidate pitch  $P$ , and a doubling threshold  $D_{th}$ , and returns the checked pitch  $P_C$ , and the corresponding correlation,  $r(P_C)$ . All fractional pitch calculations are made using the signal given to the doubling check procedure.

This procedure begins with a fractional pitch refinement around  $P$ . This produces tentative values for  $P_C$  and  $r(P_C)$ . Next, the largest value of  $k$  is found for which  $r(P_C / k) > D_{th} r(P_C)$ , where  $(P_C / k) \geq 20$  and  $k = 8, 7, \dots, 2$ .  $r(P_C / k)$  is calculated in two steps: 1) a fractional pitch refinement around  $P_C / k$ , producing  $P_k$ ; and 2) a double verification, if  $P_k < 30$ . If such a  $k$  is found, then a fractional pitch refinement around  $P_k$  is performed, producing new values for  $P_C$  and  $r(P_C)$ .

Finally, if  $P_C$  is less than 30 samples, then double verification is performed.

The following pseudo code shows the pitch double check procedure:

```

inputs: signal; P; Dth
outputs: Pc, cor_Pc

Pc, cor_Pc = frac_pitch(signal, P)
for (k=8; k>=2; k--)
    Pk = Pc/k
    if (Pk >= 20)
        Pk, cor_Pk = frac_pitch(signal, Pk)
        if (Pk < 30) cor_Pk = double_ver(Pk, cor_Pk)
        if (cor_Pk > Dth * cor_Pc)
            Pc, cor_Pc = frac_pitch(signal, Pk)
            break
        endif
    endif
endif
endfor

```

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```
if (Pc < 30) cor_Pc = double_ver(Pc, cor_Pc)
```

For inputs  $P$  and  $r(P)$ , the double verification procedure returns the smaller of  $r(P)$  and  $r(2P)$ , where  $r(2P)$  is determined by the fractional pitch procedure around  $2P$ . The use of double verification in the double check procedure provides robustness against spurious short pitch values.

**A5.2.11 Gain calculation.** The input speech signal gain is measured twice per frame using a pitch-adaptive window length. This length is identical for both gain measurements and is determined as follows. When  $Vbp_1 > 0.6$ , the window length is the shortest multiple of  $P_2$  which is longer than 120 samples. If this length exceeds 320 samples, it is divided by 2. When  $Vbp_1 \leq 0.6$ , the window length is 120 samples. The gain calculation for the first window produces  $G_1$  and is centered 90 samples before the last sample in the current frame. The calculation for the second window produces  $G_2$  and is centered on the last sample in the current frame. The gain is the RMS value, measured in dB, of the signal in the window,  $s_n$ :

$$G_1 = 10 \log_{10} \left( 0.01 + \frac{1}{L} \sum_{n=1}^L s_n^2 \right), \text{ EQUATION A-6,}$$

where  $L$  is the window length. The 0.01 term prevents the log argument from going too close to zero. If a gain measurement is less than 0.0, it is clamped to 0.0. The gain measurement assumes that the input signal range is -32768 to 32767 (see 5.2).

**A5.2.12 Average pitch update.** The long-term average pitch,  $P_{avg}$ , is updated with a simple smoothing procedure. If  $r(P_3) > 0.8$  and  $G_2 > 30\text{dB}$ , then  $P_3$  is placed into a buffer containing the three most recent strong pitch values,  $p_i$ ,  $i = 1, 2, 3$ . Otherwise, all three pitch values in the buffer are moved toward a default pitch,  $P_{default} = 50$  samples, according to:

$$p_i = 0.95p_i + 0.05P_{default}, i = 1, 2, 3, \text{ EQUATION A-7.}$$

The average pitch is then updated as the median of the three values in the buffer.  $P_{avg}$  is used in the final pitch calculation (see A5.2.9).

**A5.2.13 Quantization of prediction coefficients.** First, the linear prediction coefficients  $a_i$ ,  $i = 1, 2, \dots, 10$ , are converted into LSFs. Next, a process which forces the LSF components to be in ascending order with a minimum separation of 50 Hz is performed. This process begins by checking all adjacent pairs of the LSF components and swapping any pair not in ascending order. This step is repeated as many as ten times, if necessary. The minimum separation criterion is then applied by correcting each pair,  $f_i$  and  $f_{i+1}$ , for which  $d = f_{i+1} - f_i$  is less than 50 Hz,  $\Delta_{min}$ , as shown in the following pseudo code. The LSF components and frequency-related constants are in Hertz; scaling in other implementations may differ. The minimum separation process is repeated ten times.

```
Dmin = 50
for (I=1; I<10; I++)
    d = f[I+1] - f[I]
```

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```

if (d < dmin)
  s1 = s2 = (dmin-d)/2
  if (I == 1 and f[I] < dmin) s1 = f[I]/2
  else if (I > 1)
    tmp = f[I] - f[I-1]
    if (tmp < dmin) s1 = 0
    else if (tmp < 2*dmin) s1 = (tmp-dmin)/2
  endif
  if (I == 9 and f[I+1] > 4000-dmin) s2 = (4000-f[I+1])/2
  else if (I < 9)
    tmp = f[I+2] - f[I+1]
    if (tmp < dmin) s2 = 0
    else if (tmp < 2*dmin) s2 = (tmp-dmin)/2
  endif
  f[I+1] = f[I+1] + s2
endif
endfor

```

The resulting LSF vector,  $f$ , is then quantized using a MSVQ. The MSVQ codebook consists of four stages of 128, 64, 64, and 64 levels respectively. The quantized vector,  $\hat{f}$ , is the sum of the vectors selected by the search process, with one vector selected from each stage. The MSVQ search finds the codebook vector which minimizes the square of the weighted Euclidean distance,  $d^2$ , between the unquantized and quantized LSF vectors:

$$d^2(f, \hat{f}) = \sum_{i=1}^{10} w_i (f_i - \hat{f}_i)^2, \text{ where } w_i = \begin{cases} P(f_i)^{0.3}, & 1 \leq i \leq 8 \\ 0.64P(f_i)^{0.3}, & i = 9 \\ 0.16P(f_i)^{0.3}, & i = 10 \end{cases}, \text{ EQUATION A-8,}$$

$f_i$  is the  $i^{\text{th}}$  component of the unquantized LSF vector, and  $P(f_i)$  is the inverse prediction filter power spectrum evaluated at frequency  $f_i$ . The search procedure is an M-best approximation to a full search, in which the M=8 best code vectors from each stage are saved for use with the next stage. The process to ensure ascending order and minimum separation (described in the first part of this section) is then applied to the quantized LSF vector. The resulting vector is used in the Fourier magnitude calculation (see A5.2.17).

**A5.2.14 Pitch quantization.** The final pitch value,  $P_3$ , is quantized on a logarithmic scale with a 99-level uniform quantizer ranging from 20 to 160 samples. These pitch values are then mapped to a 7-bit codeword using a look-up table, as shown in section 5.3.1. The all-zero codeword represents the unvoiced state, and is sent if  $Vbp_1 \leq 0.6$ . All 28 codewords with Hamming weight of 1 or 2 are reserved for error protection. The uniform quantizer details are described in 5.3.7.

**A5.2.15 Gain quantization.** The two gain values are quantized as follows.  $G_2$  is quantized with a 5-bit uniform quantizer ranging from 10 to 77 dB.  $G_1$  is quantized to 3 bits using the following adaptive algorithm. If  $G_2$  for the current frame is within 5 dB of  $G_2$  for the previous frame, and  $G_1$  is within 3 dB of the average of the  $G_2$  values for the current and previous frames, then the frame is steady-state and a special code (all zero) is

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sent to indicate that the decoder should set  $G_1$  to the mean of the  $G_2$  values for the current and previous frames. Otherwise, the frame represents a transition and  $G_1$  is quantized with a 7-level uniform quantizer ranging from 6 dB below the minimum of the  $G_2$  values for the current and previous frames to 6 dB above the maximum of those  $G_2$  values. The quantizer range is clamped to 10 and 77 dB. The uniform quantizer details are described in 5.3.7. Pseudo code for the adaptive quantization of  $G_1$  is shown below.

```

If (|G2 - G2p| < 5.0 and |G1 - 0.5 * (G2 + G2p)| < 3.0)
    quantizer_index = 0
else
    gain_max = max(G2p, G2) + 6.0
    gain_min = min(G2p, G2) - 6.0
    if (gain_min < 10.0) gain_min = 10.0
    if (gain_max > 77.0) gain_max = 77.0
    quantizer_index values 1 to 7 are determined by quantizing G1 with a 7-level,
    uniform quantizer ranging from gain_min to gain_max
endif

```

A5.2.16 Bandpass voicing quantization. When  $Vbp_1 \leq 0.6$  (unvoiced), the remaining voicing strengths,  $Vbp_i, i = 2,3,4,5$ , are quantized to 0. When  $Vbp_1 > 0.6$ , the remaining voicing strengths are quantized to 1 if their value exceeds 0.6, and quantized to 0 otherwise. There is one exception. If the quantized values of  $Vbp_i, i = 2,3,4,5$  are 0001, respectively, then  $Vbp_5$  is quantized to 0.

A5.2.17 Fourier magnitude calculation and quantization. This analysis measures the Fourier magnitudes of the first 10 pitch harmonics of the prediction residual generated by the quantized prediction coefficients. It uses a 512-point Fast Fourier Transform (FFT) of a 200 sample window centered at the end of the frame. First, a set of quantized predictor coefficients is calculated from the quantized LSF vector (see A5.2.13). Then the residual window is generated using the quantized prediction coefficients. Next, a 200 sample Hamming window is applied, the signal is zero-padded to 512 points, and the complex FFT is performed. Finally, the complex FFT output is transformed into magnitudes, and the harmonics are found with a spectral peak-picking algorithm.

The peak-picker finds the maximum within a width of  $512 / \hat{P}_3$  frequency samples centered around the initial estimate for each pitch harmonic, where  $\hat{P}_3$  is the quantized pitch. This width is truncated to an integer. The initial estimate for the location of the  $I^{\text{th}}$  harmonic is  $512i / \hat{P}_3$ . The number of harmonic magnitudes searched for is limited to the smaller of 10 or  $\hat{P}_3 / 4$ . These magnitudes are then normalized to have an RMS value of 1.0. If fewer than 10 harmonics are found, the remaining magnitudes are set to 1.0.

The 10 magnitudes are quantized with an 8-bit vector quantizer. The codebook is searched using a perceptually weighted Euclidean distance, with fixed weights that emphasize low frequencies over higher frequencies. The weights are given by:

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$$w_i = \left[ \frac{117}{25 + 75 \left( 1 + 1.4 \left( \frac{f_i}{1000} \right)^2 \right)^{0.69}} \right]^2, i = 1, 2, \dots, 10, \text{ EQUATION A-9,}$$

where  $f_i = 8000i / 60$  is the frequency in Hz corresponding to the  $i^{\text{th}}$  harmonic for a default pitch period of 60 samples. The weights are applied to the squared difference between the input Fourier magnitudes and the codebook values.

**A5.2.18 Error protection and bit packing.** Table II in 5.5.2 shows the bit allocation for the MELP coder. To improve performance in channel errors, the unused coder parameters for the unvoiced mode are replaced with forward error correction. Three Hamming (7,4) codes and one Hamming (8,4) code are used. The (7,4) code corrects single bit-errors, while the (8,4) code in addition detects double bit-errors. The (8,4) code is applied to the 4 most significant bits (MSBs) of the first MSVQ index, and the 4 parity bits are written over the bandpass voicing. The remaining 3 bits of the first MSVQ index along with a reserved bit (set to zero), are covered by a (7,4) code with the resulting 3 parity bits written to the MSBs of the Fourier series VQ index. The 4 MSBs of the  $G_2$  codeword are protected with 3 parity bits which are written to the next 3 bits of the Fourier magnitudes. Finally, the LSB of the second gain index and the 3 bit  $G_1$  codeword are protected with 3 parity bits written to the 2 LSBs of the Fourier magnitudes and the aperiodic flag.

The bit transmission order is given in 5.5.3.

**A.5.3 Decoder.** The channel data is decoded by performing the following steps in the order given.

**A5.3.1 Bit unpacking and error correction.** The received bits are unpacked from the channel and assembled into the parameter codewords. Parameter decoding is different for voiced and unvoiced modes. The pitch is decoded first, since it contains the mode information. If the pitch code is all-zero or has only one bit set, then the unvoiced mode is used. If two bits are set, a frame erasure is indicated. Otherwise, the pitch value is decoded and the voiced mode is used.

In the unvoiced mode, the (8,4) Hamming code is decoded to correct single bit errors and detect double errors. If an uncorrectable error is detected, a frame erasure is indicated. Otherwise, the (7,4) Hamming codes are decoded, correcting single errors but without double error detection.

If any erasure is detected in the current frame, by the Hamming code, by the pitch code, or directly signaled from the channel, then a frame repeat mechanism is implemented. All of the parameters for the current frame are replaced with the parameters from the previous frame. In addition, the first gain term is set equal to the second gain term so that no gain transitions are allowed.

If an erasure is not indicated, the remaining parameters are decoded. The LSFs are checked for ascending order and minimum separation as described in A5.2.13. In the unvoiced mode, default parameter values are used for the pitch, jitter, bandpass voicing, and Fourier magnitudes. The pitch value is set to 50 samples, the jitter is set to

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25%, all of the bandpass voicing strengths are set to 0, and the Fourier magnitudes are set to 1. In the voiced mode,  $V_{bp_1}$  is set to 1; jitter is set to 25% if the aperiodic flag is a 1; otherwise jitter is set to 0%. The bandpass voicing strength for the upper four bands is set to 1 if the corresponding bit is a 1; otherwise the voicing strength is set to 0. There is one exception. If 0001 is received for  $V_{bp_i}$ ,  $i = 2, 3, 4, 5$ , respectively, then  $V_{bp_5}$  is set to 0.

When the special all-zero code for the first gain parameter,  $G_1$ , is received, some errors in the second gain parameter,  $G_2$ , can be detected and corrected. This correction process provides improved performance in channel errors. The decoding for the two gain parameters is shown in the following pseudo code.

```

Inputs: G1_index, G2_index
outputs: G1, G2
internal: G2p, G2p_error

G2 = decode(G2_index)           32 levels; range: 10 to 77 dB
if (G1_index == 0)              special G1 code: use mean of G2 and G2p
    if (|G2 - G2p| > 5)          G2_index probably in error
        if (G2p_error == 0)     G2p is correct
            G2 = G2p            replace the erroneous G2 with past value
        endif
        G2p_error = 1
    else G2_index probably correct
        G2p_error = 0
    endif
    G1 = 0.5 * (G2 + G2p)        mean of G2 and G2p
else
    G1 = decode(G1_index)       7 levels; range: min(G2,G2p)-6 to max(G2,G2p)+6
    G2p_error = 0               (above range is clamped to 10 to 77 dB)
endif
G2p = G2                        save for use as past value

```

**A5.3.2 Noise attenuation.** For quiet input signals, a small amount of gain attenuation is applied to both decoded gain parameters using a power subtraction rule. This attenuation is a simplified, frequency invariant case of the Smoothed Spectral Subtraction noise suppression method.

Before determining the attenuation for the first gain term,  $G_1$ , a background noise estimate,  $G_n$ , is updated as follows. If  $G_1 > G_n + C_{up}$  then  $G_n = G_n + C_{up}$ . If  $G_1 < G_n - C_{down}$  then  $G_n = G_n - C_{down}$ . Otherwise,  $G_n = G_1$ .  $C_{up} = 0.0337435$  and  $C_{down} = 0.135418$ , so that the noise estimator moves up by 3 dB per second and down by 12 dB per second for the gain update rate of 88.9 updates per second. The noise estimate is clamped between 10 and 80. Noise estimation is disabled for repeated frames to prevent repeated attenuation. The background noise estimate is also used in the adaptive spectral enhancement calculation (see A5.3.5).

Gain  $G_1$  is then modified by subtracting a (positive) correction term,  $G_{att}$ , given in dB by

$$G_{att} = -10 \log_{10} \left( 1 - 10^{0.1[G_n + 3 - G_1]} \right), \text{EQUATION A-10,}$$

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where  $G_n$  is the background noise estimate (in dB), and  $G_1$  is the first gain term (in dB). The correction is clamped to a maximum value of 6 dB to avoid fluctuations and signal distortion. To ensure that the attenuation is applied only to quiet signals, the  $G_n$  value as used in equation A-10 is clamped at an upper limit of 20 dB.

The noise estimation and gain modification steps are then repeated for the second gain term,  $G_2$ . Noise estimation and gain attenuation are disabled for repeated frames.

**A5.3.3 Parameter interpolation.** All MELP synthesis parameters are interpolated pitch-synchronously for each synthesized pitch period. The interpolated parameters are the gain (in dB), LSFs, pitch, jitter, Fourier magnitudes, pulse and noise coefficients for mixed excitation, and spectral tilt coefficient for the adaptive spectral enhancement filter. Gain is linearly interpolated between the second gain of the prior frame,  $G_{2p}$ , and the first gain of the current frame,  $G_1$ , if the starting point,  $t_0, t_0 = 0, 1, \dots, 179$ , of the new pitch period is less than 90; otherwise, gain is interpolated between  $G_1$  and  $G_2$ . Normally, the other parameters are linearly interpolated between the past and current frame values. The interpolation factor,  $int$ , for these parameters is based on the starting point of the new pitch period:

$$int = t_0 / 180, \text{ EQUATION A-11.}$$

There are two exceptions to this interpolation procedure. First, if there is an onset with a high pitch frequency, pitch interpolation is disabled and the new pitch is immediately used. This condition is met when  $G_1$  is more than 6 dB greater than  $G_{2p}$  and the current frame's pitch period is less than half the prior frame's pitch period. The second exception also involves a gain onset. If  $G_2$  differs from  $G_{2p}$  by more than 6 dB, then the LSF's, spectral tilt, and pitch are interpolated using the interpolated gain trajectory as a basis, since the gain is transmitted twice per frame and has a more accurate interpolation path. In this case, the interpolation factor is given by

$$int = \frac{G_{int} - G_{2p}}{G_2 - G_{2p}}, \text{ EQUATION A-12,}$$

where  $G_{int}$  is the interpolated gain. This interpolation factor is then clamped between 0 and 1.

**A5.3.4 Mixed excitation generation.** The mixed excitation is generated as the sum of the filtered pulse and noise excitations. The pulse excitation,  $e_p(n)$ ,  $n = 0, 1, \dots, T-1$ , is computed using an inverse Discrete Fourier Transform of one pitch period in length:

$$e_p(n) = \frac{1}{T} \sum_{k=0}^{T-1} M(k) e^{j2\pi nk / T}, \text{ EQUATION A-13.}$$

The pitch period,  $T$ , is the interpolated pitch value plus the jitter times the pitch, where the jitter is the interpolated jitter strength times the output of a uniform random number generator between -1 and 1. This pitch period is rounded to the nearest integer and clamped between 20 and 160. All of the phases for the pulse excitation are set to zero, hence  $M(k)$  is real. Since  $e_p(n)$  is real, the magnitudes obey:

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$$M(T-k) = M(k), k = 1, 2, \dots, L, \text{ EQUATION A-14,}$$

where  $L = T/2$  if  $T$  is even, and  $L = (T-1)/2$ , if  $T$  is odd. The DC term,  $M(0)$ , is set to 0. Magnitude terms  $M(k)$ ,  $k = 1, 2, \dots, 10$ , are set to the interpolated values of the Fourier magnitudes, and any magnitudes not otherwise specified are set to 1. To prevent rapid changes at the start of the pitch period, the pulse excitation is circularly shifted by ten samples of delay so the main excitation pulse occurs at the tenth sample of the period. The pulse is then multiplied by the square root of the pitch to give a unity RMS signal, and then multiplied by 1000 to give a nominal signal level.

The noise is generated by a uniform random number generator with an RMS value of 1000, and range of -1732 to 1732.

The pulse and noise excitation signals are then filtered and summed to form the mixed excitation. The pulse filter for the current frame is given by the sum of all the bandpass filter coefficients for the voiced frequency bands, while the noise filter is given by the sum of the bandpass filter coefficients for the unvoiced bands. These filter coefficients are interpolated pitch synchronously. The bandpass filter coefficients for each of the five bands are given in table A-I.

| TABLE A-I. Filter coefficients for bandpass filter. |             |              |              |             |
|-----------------------------------------------------|-------------|--------------|--------------|-------------|
| 0-500 Hz                                            | 500-1000 Hz | 1000-2000 Hz | 2000-3000 Hz | 3000-4000Hz |
| -0.00302890                                         | -0.00249420 | -0.00231491  | 0.00231491   | 0.00554149  |
| -0.00701117                                         | -0.00282091 | 0.00990113   | 0.00990113   | -0.00981750 |
| -0.01130619                                         | 0.00257679  | 0.02086129   | -0.02086129  | 0.00856805  |
| -0.01494082                                         | 0.01451271  | -0.00000000  | 0.00000000   | -0.00000000 |
| -0.01672586                                         | 0.02868120  | -0.03086123  | 0.03086123   | -0.01267517 |
| -0.01544189                                         | 0.03621179  | -0.02180695  | -0.02180695  | 0.02162277  |
| -0.01006619                                         | 0.02784469  | 0.00769333   | -0.00769333  | -0.01841647 |
| 0.00000000                                          | 0.00000000  | -0.00000000  | -0.00000000  | 0.00000000  |
| 0.01474923                                          | -0.04079870 | -0.01127245  | 0.01127245   | 0.02698425  |
| 0.03347158                                          | -0.07849207 | 0.04726837   | 0.04726837   | -0.04686914 |
| 0.05477206                                          | -0.09392213 | 0.10106105   | -0.10106105  | 0.04150730  |
| 0.07670890                                          | -0.07451087 | -0.00000000  | 0.00000000   | -0.00000000 |
| 0.09703726                                          | -0.02211575 | -0.17904543  | 0.17904543   | -0.07353666 |
| 0.11352143                                          | 0.04567473  | -0.16031428  | -0.16031428  | 0.15896026  |
| 0.12426379                                          | 0.10232715  | 0.09497157   | -0.09497157  | -0.22734513 |
| 0.12799355                                          | 0.12432599  | 0.25562154   | 0.25562154   | 0.25346255  |

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| <b>TABLE A-I. Filter coefficients for bandpass filter - Continued.</b> |                    |                     |                     |                    |
|------------------------------------------------------------------------|--------------------|---------------------|---------------------|--------------------|
| <b>0-500 Hz</b>                                                        | <b>500-1000 Hz</b> | <b>1000-2000 Hz</b> | <b>2000-3000 Hz</b> | <b>3000-4000Hz</b> |
| 0.12426379                                                             | 0.10232715         | 0.09497157          | -0.09497157         | -0.22734513        |
| 0.11352143                                                             | 0.04567473         | -0.16031428         | -0.16031428         | 0.15896026         |
| 0.09703726                                                             | -0.02211575        | -0.17904543         | 0.17904543          | -0.07353666        |
| 0.07670890                                                             | -0.07451087        | -0.00000000         | 0.00000000          | -0.00000000        |
| 0.05477206                                                             | -0.09392213        | 0.10106105          | -0.10106105         | 0.04150730         |
| 0.03347158                                                             | -0.07849207        | 0.04726837          | 0.04726837          | -0.04686914        |
| 0.01474923                                                             | -0.04079870        | -0.01127245         | 0.01127245          | 0.02698425         |
| 0.00000000                                                             | -0.00000000        | -0.00000000         | -0.00000000         | 0.00000000         |
| -0.01006619                                                            | 0.02784469         | 0.00769333          | -0.00769333         | -0.01841647        |
| -0.01544189                                                            | 0.03621179         | -0.02180695         | -0.02180695         | 0.02162277         |
| -0.01672586                                                            | 0.02868120         | -0.03086123         | 0.03086123          | -0.01267517        |
| -0.01494082                                                            | 0.01451271         | -0.00000000         | 0.00000000          | -0.00000000        |
| -0.01130619                                                            | 0.00257679         | 0.02086129          | -0.02086129         | 0.00856805         |
| -0.00701117                                                            | -0.00282091        | 0.00990113          | 0.00990113          | -0.00981750        |
| -0.00302890                                                            | -0.00249420        | -0.00231491         | 0.00231491          | 0.00554149         |

A5.3.5 Adaptive spectral enhancement. The adaptive spectral enhancement filter is applied to the mixed excitation signal. This filter is a tenth order pole/zero filter, with an additional first-order tilt compensation. Its coefficients are generated by bandwidth expansion of the linear prediction filter transfer function,  $A(z)$ , corresponding to the interpolated LSFs. The transfer function of the enhancement filter,  $H_{ase}(z)$ , is given by:

$$H_{ase}(z) = \frac{A(\alpha z^{-1})}{A(\beta z^{-1})} * (1 + \mu z^{-1}), \text{ where } \begin{matrix} \alpha = 0.5p \\ \beta = 0.8p \end{matrix}, \text{ EQUATION A-15,}$$

and tilt the coefficient  $\mu$  is first calculated as  $\max(0.5k_1, 0)$ , then interpolated, then multiplied by  $p$ , the signal probability. The first reflection coefficient,  $k_1$ , is calculated from the decoded LSFs. By the MELP predictor coefficient sign convention,  $k_1$  is usually negative for voiced spectra. The signal probability  $p$  is estimated by comparing the current interpolated gain,  $G_{int}$ , to the background noise estimate  $G_n$  using the formula:

$$p = \frac{G_{int} - G_n - 12}{18}, \text{ EQUATION A-16.}$$

This signal probability is clamped between 0 and 1.

A5.3.6 Linear prediction synthesis. The synthesis uses a direct form filter, with the coefficients corresponding to the interpolated LSFs.

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A5.3.7 Gain adjustment. Since the excitation is generated at an arbitrary level, the speech gain must be introduced to the synthesized speech. The correct scaling factor,  $S_{\text{gain}}$ , is computed for each synthesized pitch period of length  $T$  by dividing the desired RMS value ( $G_{\text{int}}$  must be converted from dB) by the RMS value of the unscaled synthetic speech signal  $\hat{s}_n$ :

$$S_{\text{gain}} = \frac{10^{G_{\text{int}}/20}}{\sqrt{\frac{1}{T} \sum_{n=1}^T \hat{s}_n^2}}, \text{ EQUATION A-19.}$$

To prevent discontinuities in the synthesized speech, this scale factor is linearly interpolated between the previous and current values for the first ten samples of the pitch period.

A5.3.8 Pulse dispersion. The pulse dispersion filter is a 65<sup>th</sup> order FIR filter derived from a spectrally-flattened triangle pulse. The coefficients are listed in table A-II.

| TABLE A-II. Filter coefficients for the pulse dispersion filter. |               |               |               |               |
|------------------------------------------------------------------|---------------|---------------|---------------|---------------|
| Samples 1-13                                                     | Samples 14-26 | Samples 27-39 | Samples 40-52 | Samples 53-65 |
| -0.17304259                                                      | 0.24325127    | 0.07343483    | 0.02968464    | 0.00019707    |
| -0.01405709                                                      | -0.01767043   | -0.00518645   | -0.01247640   | -0.02825247   |
| 0.01224406                                                       | -0.00018612   | 0.01298488    | 0.01854666    | 0.01720989    |
| 0.11364226                                                       | 0.05869485    | 0.02928440    | 0.00076184    | -0.06004292   |
| 0.00198199                                                       | -0.00327456   | -0.01989405   | -0.07749640   | -0.07076744   |
| 0.00000658                                                       | 0.00607395    | 0.01216758    | 0.01244697    | 0.00914347    |
| 0.04529633                                                       | 0.02753924    | 0.01180979    | -0.02721777   | 0.06082730    |
| -0.00092027                                                      | -0.03351673   | -0.38924775   | 0.07266098    | 0.01805528    |
| -0.00103078                                                      | 0.00602189    | 0.00720325    | 0.00472008    | -0.00318634   |
| 0.02552787                                                       | 0.01436539    | -0.01154561   | 0.03526439    | 0.03444110    |
| -0.06339257                                                      | 0.82854582    | 0.08426287    | 0.02674603    | 0.00026302    |
| -0.00122031                                                      | 0.00033165    | -0.00355720   | -0.00744038   | -0.01053809   |
| 0.01412525                                                       | -0.00360180   | 0.02151233    | 0.02582623    | 0.02165922    |

A5.3.9 Synthesis loop control. After processing each pitch period, the decoder updates  $t_0$  by adding  $T$ , the number of samples in the period just synthesized. If  $t_0 < 180$ , synthesis for the current frame continues from the parameter interpolation step (see A5.3.3). Otherwise, the decoder buffers the remainder of the current period which extends beyond the end of the current frame and subtracts 180 from  $t_0$  to produce its initial value next frame.

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## PERFORMANCE VERIFICATION

## B.1 SCOPE

B.1.1 Scope. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance. All new implementations of the MELP coder must be tested to verify that their performance meets or exceeds that of the MELP reference coder. This appendix provides guidelines for verifying the performance of a MELP implementation. Two methods of verification are presented.

## B.2 APPLICABLE DOCUMENTS

B.2.1 Government documents. Not applicable.

B.2.2 Other publications. The following documents form a part of this appendix to the extent specified.

## ANSI Standard

S3.2-1989

American National Standard Method for  
Measuring the Intelligibility of Speech  
over Communications Systems

(Applications for copies should be addressed to ANSI Customer Service, 11 West 42<sup>nd</sup> Street, New York, New York 10036, USA)

## INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

"IEEE Recommended Practice for Speech Quality Measurements"  
by IEEE Subcommittee on Subjective Measurements, Transactions on Audio and  
Electroacoustics, Vol. 17, 1969, pp. 227-246

(Applications for copies should be addressed to IEEE Customer Service, 445 Hoes Lane, P.O. Box 1331 Piscataway, New Jersey 08855-1331, USA)

B.2.3 Order of precedence. In the event of a conflict between the text of this standard and the references stated herein, the text of this standard should take precedence.

## B.3 DEFINITIONS

B.3.1 Terms. Terms used in this appendix are defined in section 3.1 of the standard or as follows.

B3.1.1 A/B test. An A/B test is a direct paired forced choice comparison test that is used to assess the quality of one voice coder against another voice coder.

B.3.2 Acronyms used in this appendix. Acronyms used in this section are either defined in 3.2 of the standard or as follows.

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ANSI - American National Standards Institute

DRT - Diagnostic Rhyme Test

PCM - Pulse Code Modulation

## B.4 GENERAL REQUIREMENTS

B.4.1 General. New implementations of MELP must be verified to assure that their performance meets or exceeds the performance of the MELP reference coder. A new implementation must also meet the same performance standards when interoperating with the MELP reference coder. Testing is accomplished through formal evaluation of intelligibility and quality or by showing bit equivalence between the new implementation and a verified MELP implementation. Both verification methods evaluate an implementation over an extensive set of conditions.

## B.5 DETAILED REQUIREMENTS

B.5.1 Formal evaluation. The intelligibility of a new implementation is evaluated using the DRT. DRT performance thresholds have been set and are discussed later in this section. Quality is evaluated using a direct paired forced choice comparison test, i.e., an A/B test. Performance thresholds for the quality tests have also been set and are based on the percent preference for the new implementation over the MELP reference coder. The Federal Standard CELP coder is also included in the quality test to broaden the context of the test and to include a known difference from MELP. Table B-I summarizes the coder configurations which are evaluated for intelligibility and quality. In table B-I, the "Implementation → MELP Reference" and "MELP Reference → Implementation" cases are "cross-coder" configurations which test interoperability. Table B-II shows the intelligibility and quality test conditions. Six talkers (3 male, 3 female) are used for each condition.

**TABLE B-I. Tested coder configurations.**

| <b>Intelligibility (Encoder → Decoder)</b> | <b>Quality (Encoder → Decoder)</b> |
|--------------------------------------------|------------------------------------|
| Implementation → Implementation            | Implementation → Implementation    |
| Implementation → MELP Reference            | Implementation → MELP Reference    |
| MELP Reference → Implementation            | MELP Reference → Implementation    |
|                                            | MELP Reference → MELP Reference    |
|                                            | CELP → CELP                        |

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| <b>TABLE B-II. Intelligibility and quality test conditions.</b> |                                          |
|-----------------------------------------------------------------|------------------------------------------|
| <b>Intelligibility Condition (microphone)</b>                   | <b>Quality Condition (microphone)</b>    |
| Quiet (Dynamic)                                                 | Quiet (Dynamic)                          |
| Quiet (H250)                                                    | Quiet (H250)                             |
| Office (STU-III)                                                | Office (STU-III)                         |
| E3A AWACS (R215)                                                | E3A AWACS (R215)                         |
| HMMWV (H250)                                                    | HMMWV (H250)                             |
| MCE Field Shelter (M87)                                         | MCE Field Shelter (M87)                  |
| Car (STU-III cellular)                                          | Car (STU-III cellular)                   |
| 0.5% Block Error Rate in Quiet (Dynamic)                        | 0.5% Block Error Rate in Quiet (Dynamic) |
| 1.0% Bit Error Rate in Quiet (Dynamic)                          | 1.0% Bit Error Rate in Quiet (Dynamic)   |
| CVSD → Coder in Quiet (Dynamic)                                 | CVSD → Coder in Quiet (Dynamic)          |
| CVSD → Coder → CVSD in Quiet (Dynamic)                          | CVSD → Coder → CVSD in Quiet (Dynamic)   |
| M2 Bradley (M138)                                               |                                          |
| CH47 helicopter (M87)                                           |                                          |
| F-15 jet (M101)                                                 |                                          |
| P3C Orion plane (EV985)                                         |                                          |

B5.1.1 Intelligibility tests. The DRT will be performed in accordance with ANSI standard S3.2-1989 and will be scored with eight listeners. The combined talker score determined by the test lab must meet or exceed the corresponding threshold score for each condition and for the weighted combination of all conditions. Table B-III shows the weight and threshold score for each condition. The implementation's combined score is calculated by multiplying its individual score for each condition by the corresponding weight given in table B-III. The results are then summed to produce the combined result. This process is used for each of the three coder configurations evaluated for intelligibility. The threshold score for each condition is based on a one-tail 99.5% confidence interval; the combined threshold score is based on a one-tail 99% confidence interval.

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| <b>TABLE B-III. Weights and thresholds for intelligibility conditions.</b> |               |                        |
|----------------------------------------------------------------------------|---------------|------------------------|
| <b>Intelligibility condition (microphone)</b>                              | <b>Weight</b> | <b>Threshold score</b> |
| Quiet (Dynamic)                                                            | 0.100         | 90.80                  |
| Quiet (H250)                                                               | 0.100         | 88.77                  |
| Office (STU-III)                                                           | 0.0667        | 89.34                  |
| E3A AWACS (R215)                                                           | 0.0667        | 85.10                  |
| HMMWV (H250)                                                               | 0.0667        | 61.88                  |
| MCE Field Shelter (M87)                                                    | 0.0667        | 87.28                  |
| Car (STU-III cellular)                                                     | 0.0667        | 82.48                  |
| M2 Bradley (M138)                                                          | 0.0667        | 61.33                  |
| CH47 helicopter (M87)                                                      | 0.0667        | 63.27                  |
| F-15 jet (M101)                                                            | 0.0667        | 75.06                  |
| P3C Orion plane (EV985)                                                    | 0.0667        | 82.98                  |
| 0.5% Block Error Rate in Quiet (Dynamic)                                   | 0.050         | 89.58                  |
| 1.0% Bit Error Rate in Quiet (Dynamic)                                     | 0.050         | 87.47                  |
| CVSD → Coder in Quiet (Dynamic)                                            | 0.050         | 84.07                  |
| CVSD → Coder → CVSD in Quiet (Dynamic)                                     | 0.050         | 81.33                  |
| Combined (weighted)                                                        | 1.000         | 82.634                 |

B5.1.2 Quality tests. The quality of the three coder configurations involving the new implementation (see table B-I) is compared with the quality of the MELP reference coder using a direct paired forced choice comparison (A/B) test. Quality is measured in each condition by the percent preference taken over all talkers and listeners. For each coder configuration involving the new implementation, the percent preferred must meet or exceed the threshold for each condition and for the weighted combination of all conditions.

The threshold for individual conditions is 45.84%, i.e., each coder configuration involving the new implementation must have a preference percentage of 45.84% or more in each condition. This threshold is equivalent to saying that the new implementation must be selected at least 440 times out of the 960 comparisons between the new implementation and the reference in each condition. The threshold is based on a one-tail 99.5% confidence interval.

The combined percent preferred is calculated for each of the three coder configurations by summing the weighted individual condition percentages using the weights listed in table B-IV. The combined threshold percentage is 48.71%, i.e., each of the three coder configurations involving the new implementation must have a combined preference percentage of 48.71% or more. The combined threshold is based on a one-tail 99% confidence interval.

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| <b>TABLE B-IV. Weights for quality conditions.</b> |               |
|----------------------------------------------------|---------------|
| <b>Quality condition (microphone)</b>              | <b>Weight</b> |
| Quiet (Dynamic)                                    | 0.245         |
| Quiet (H250)                                       | 0.105         |
| Office (STU-III)                                   | 0.070         |
| E3A AWACS (R215)                                   | 0.070         |
| HMMWV (H250)                                       | 0.070         |
| MCE Field Shelter (M87)                            | 0.070         |
| Car (STU-III cellular)                             | 0.070         |
| 0.5% Block Error Rate in Quiet (Dynamic)           | 0.075         |
| 1.0% Bit Error Rate in Quiet (Dynamic)             | 0.075         |
| CVSD → Coder in Quiet (Dynamic)                    | 0.075         |
| CVSD → Coder → CVSD in Quiet (Dynamic)             | 0.075         |

**B.5.2 Bit equivalence.** A lower cost alternative for implementation verification is accomplished by demonstrating that the new implementation is bit equivalent to a verified MELP implementation. Bit equivalent means that given the same input, the new implementation's encoder produces the same bitstream as produced by the encoder of the verified MELP implementation. Also given the same bitstream, the new implementation's decoder produces the same 16 bit PCM samples as produced by the decoder of the verified MELP implementation. This equivalence must be shown for all intelligibility material and quality material used in the formal evaluation.

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### CONCLUDING MATERIAL

#### Custodians:

Army – CR  
Navy – EC  
Air Force – 02  
NSA – NS

#### Preparing Activity:

NSA – NS  
  
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#### Review Activities:

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Air Force – 11, 13, 19, 93  
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3. DOCUMENT TITLE ANALOG-TO-DIGITAL CONVERSION OF VOICE BY 2,4000 BIT/SECOND MIXED EXCITATION LINEAR PREDICTION (MELP)

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